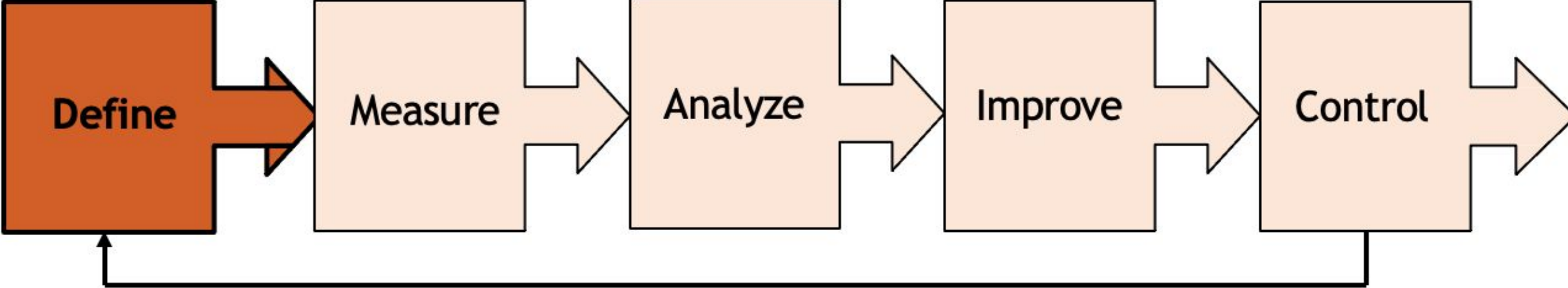


Lean Six Sigma Case Study - Gentech

Project Team #381 - The BizIntelysts

Kashyap Mehta, Ji-Lun Mo, Ning Wei, Ananya Jain



GENERAL PROJECT INFORMATION

PROJECT NAME		PROJECT MANAGER	PROJECT SPONSOR
Efficiency Improvement Proposal Creation Process		Jeff Hugh, LSS Black Belt	Grace Monroe, VP (Supply Chain Ops)
ORGANIZATIONAL UNIT			
Gentech			
GREEN BELTS ASSIGNED			
Bid Support Staff, Seller, Proposal Support Manager, Product Design, Brand Manager, Support Manager			
BLACK BELTS ASSIGNED			
Jeff Hugh		BASELINE 31.6 DAYS	TIMELINE 6 MONTHS
PROJECT TEAM	- Kashyap Mehta - Ananya Jain - Ji Lun Mo - Ning Wei	REVENUE LOSS \$ 2.49 Billion	DEFECT CYCLE TIME > 35 DAYS

PROBLEM STATEMENT:

Over the last three years, 28% of Gentech's cycle time is over 35 days, resulting in 18% reduction in revenue in the past two years.

GOAL STATEMENT:

Reduce cycle time of operations by 15% to gain competitive advantage from efficiency.

CUSTOMER:

Who: Sellers (Internal), Client (External)

Voice of Customer

Seller:

1. Difficult to identify the correct agent to route
2. BSS lack of knowledge and inexperience

BSS:

1. Redundant approvals
2. Product pricing take too long
3. Missing required information from sellers
4. Reconfiguration takes too long, seller keeps changing

CUSTOMER - MAJOR ISSUE

1. Wrong BSS Agents
2. Missing Information from sellers
3. Reconfiguration
4. Redundant approvals

Seller Feed back

Count

Incorrect configuration	7499
Incomplete requirements	3791
too much churn	3720
BSS inexperienced	7415
Errors in configuration	3796
Responses rate is too slow	3769
Too many errors by BSS	3735
Difficult to identify BSS agent	11383
pricing is not competitive	3754
BSS lacks domain knowledge	7458
ERP system doesn't work well	3677
Terms and conditions had errors	3795
Proposal takes too long	7422
Submission process is tedious	3786

BSS Feedback

Count

Missing Customer address	6516
Missing seller information	6877
Documentation provided late	7504
Error in address provided	7151
Configuration changed	4153
Incomplete information	7592
Bid manager on vacation	4589
Multiple quote submissions	4081
customer not in database	2418
Pricing approval takes too long	4735
Missing configuration details	7915
Multiple submissions	4137
System outage	2088
Brand approval takes too long	5244

DELIVERABLES & BENEFITS

EXPECTED DELIVERABLES	• A detailed map of the current proposal creation process. • A comprehensive report identifying inefficiencies and potential areas for improvement. • A new, optimized proposal creation process design. • Implementation plan for the new process, including a pilot test. • Training materials for all affected employees.
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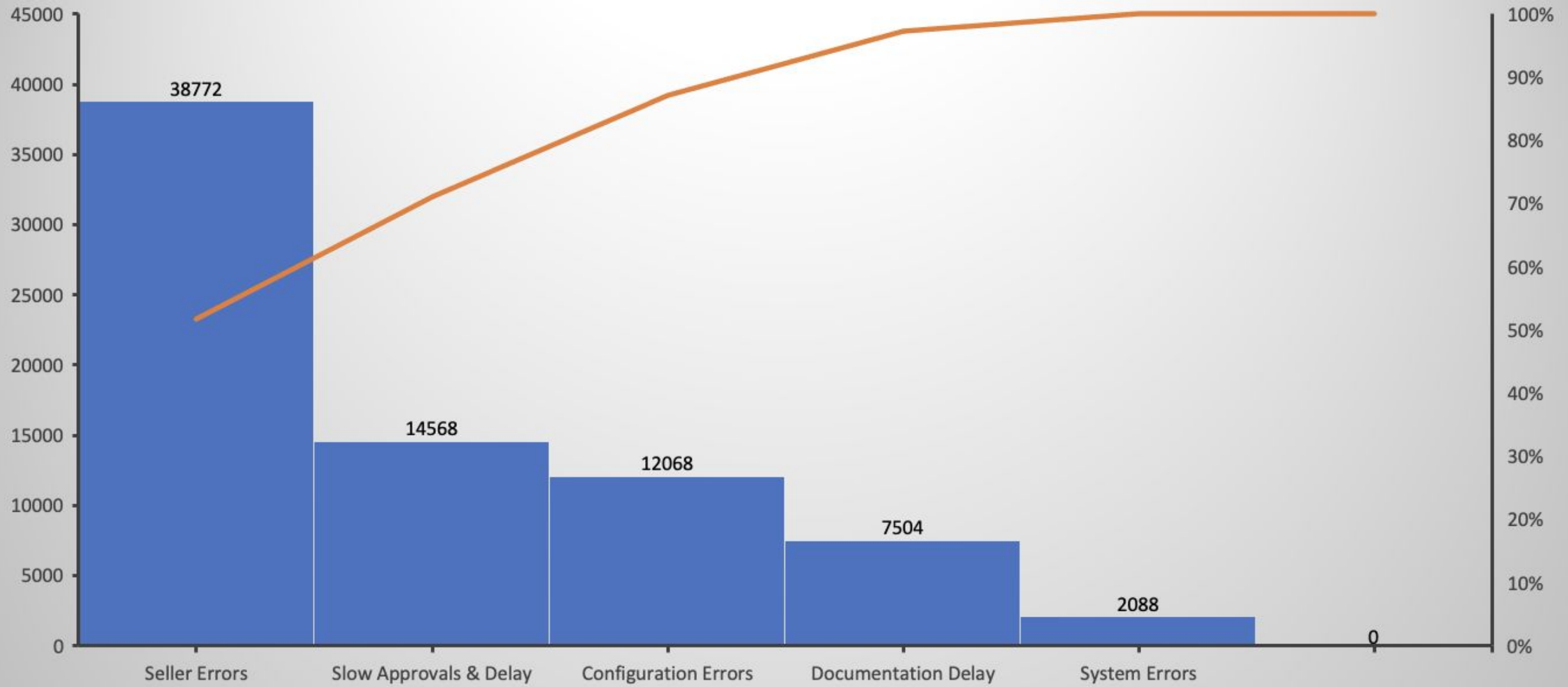
EXPECTED BENEFITS

BSS Satisfaction	Seller Satisfaction	Cycle Time Reduction	Process Improvement	Increased Efficiency
Enhanced Accuracy	Pricing Tool Automation	Parallel Processing	Gain Competitive Advantage	

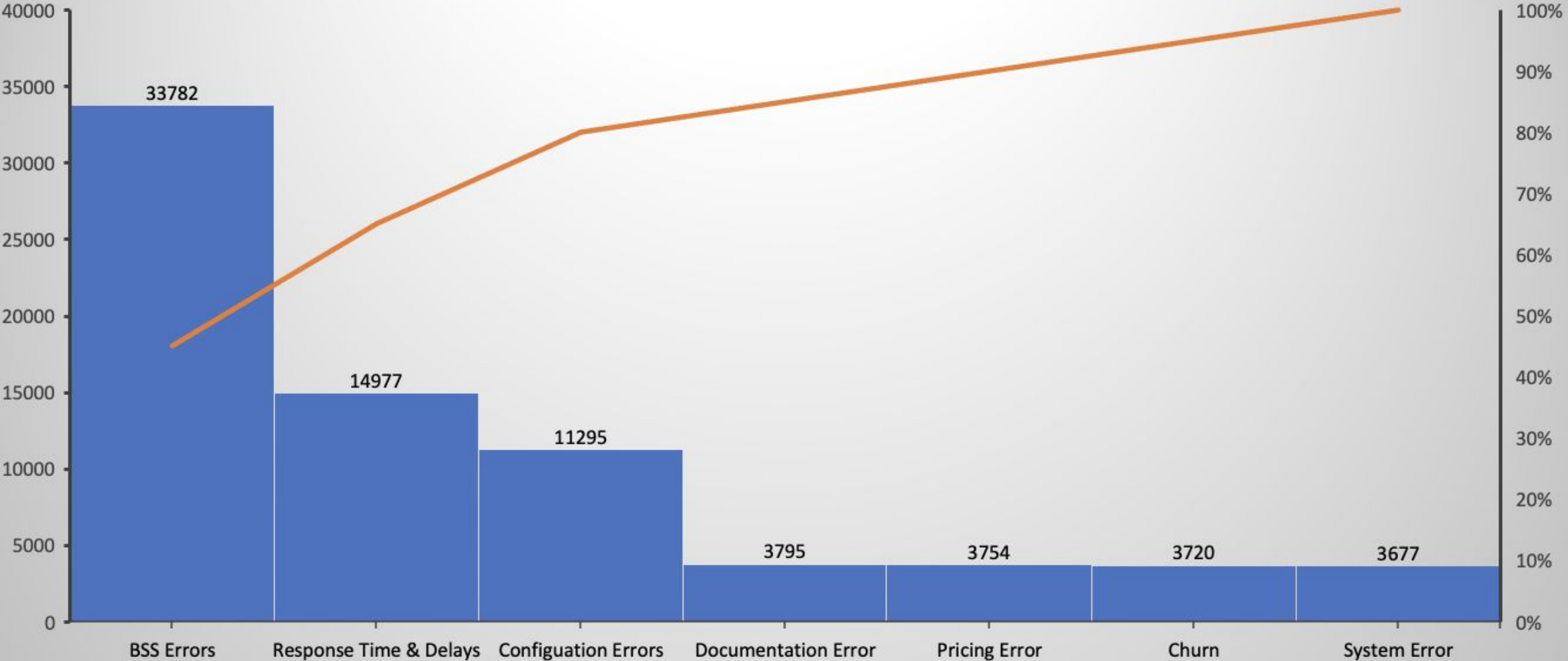
PROJECT SCOPE

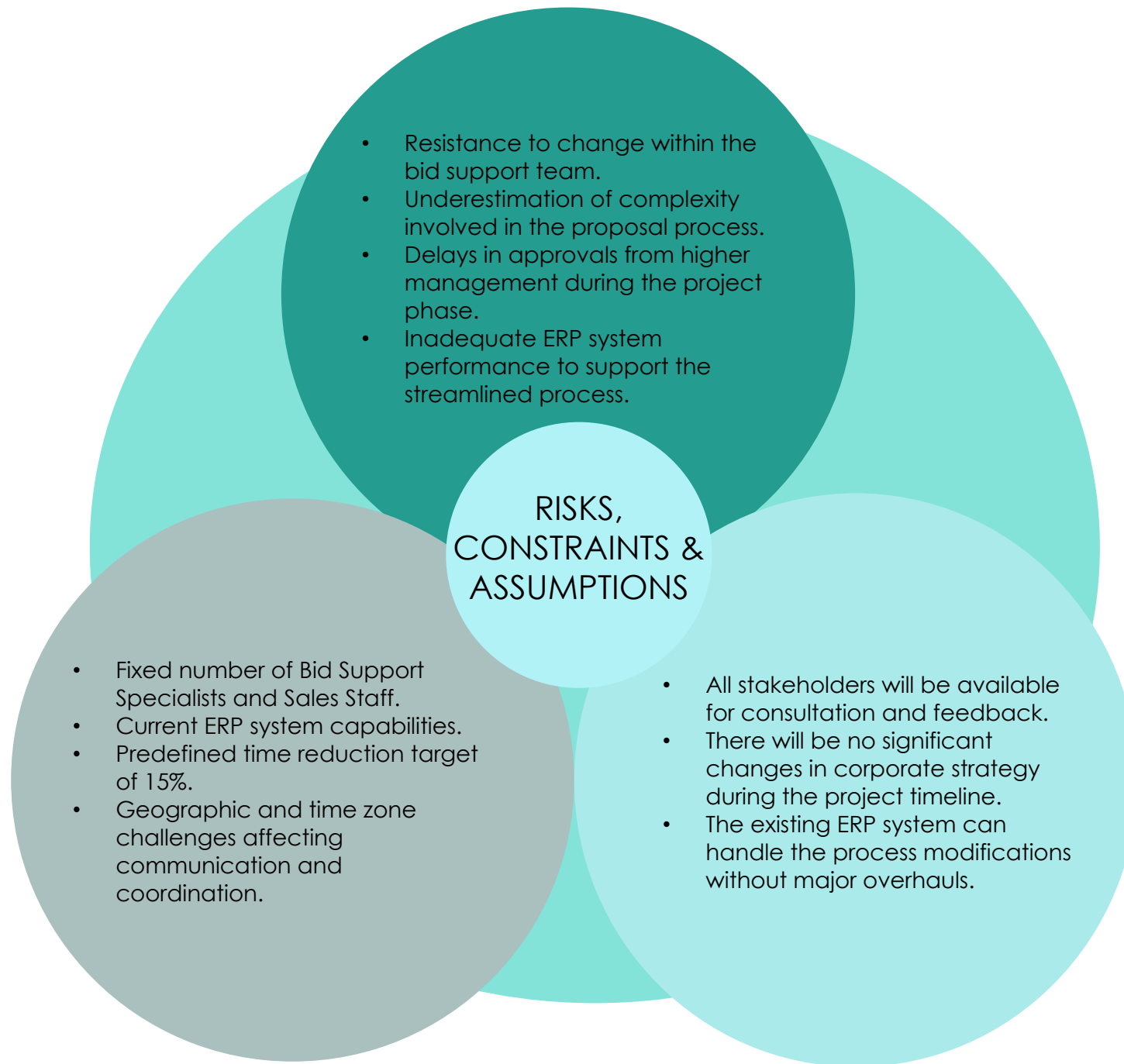
WITHIN SCOPE	Streamlining and standardization of processes for proposal creation, including bid support coordination, product design communication, and pricing approval for all Gentech brands
OUTSIDE OF SCOPE	Changes to the sales strategies, product development, customer service delivery, and other downstream processes post-proposal submission

BSS FEEDBACK



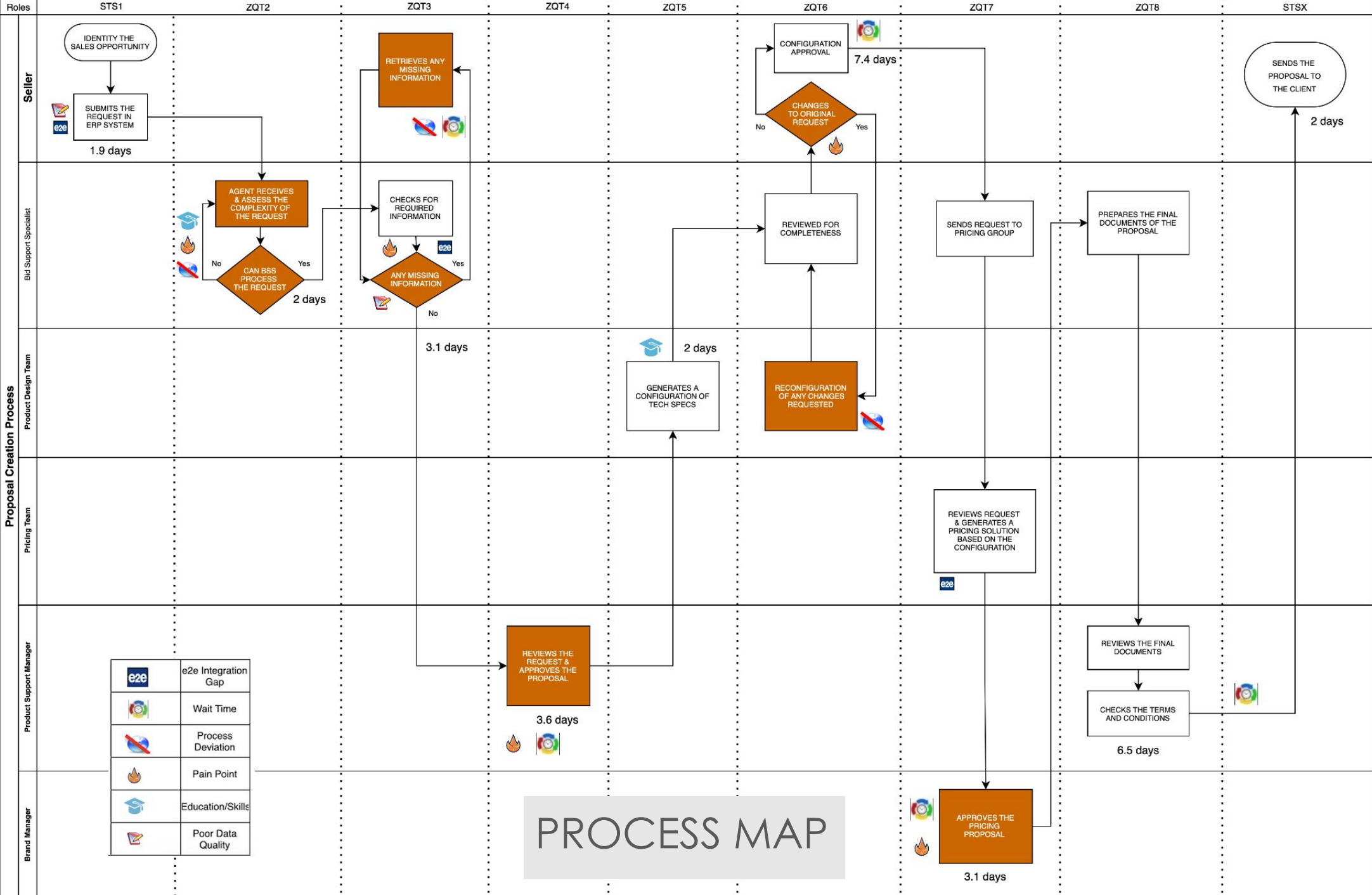
SELLER FEEDBACK



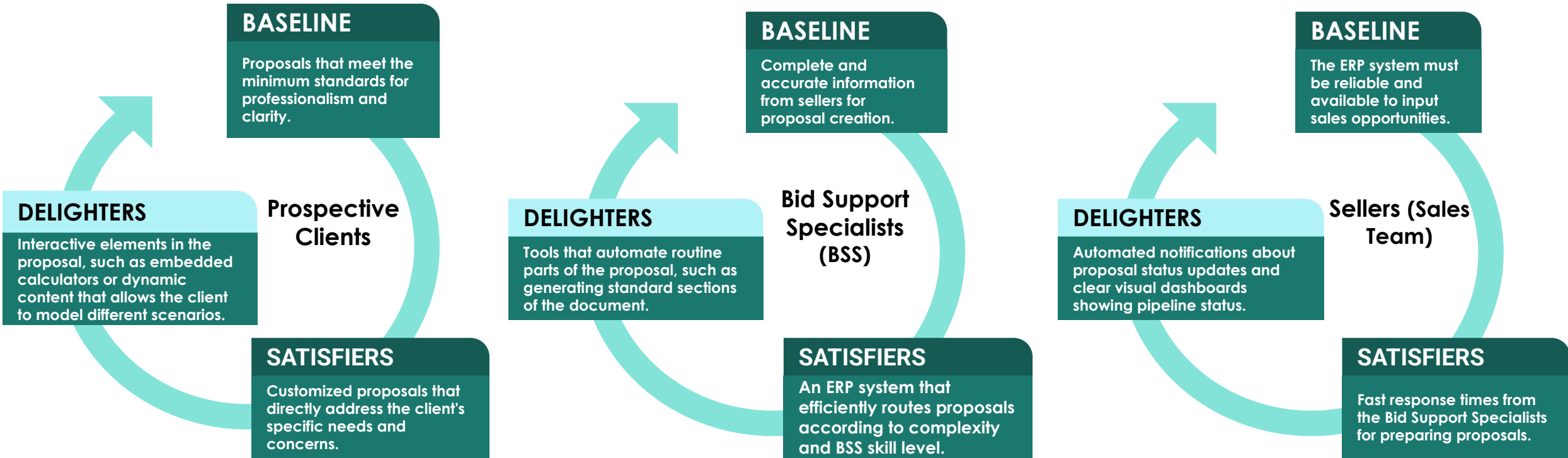


STAKEHOLDER ANALYSIS:

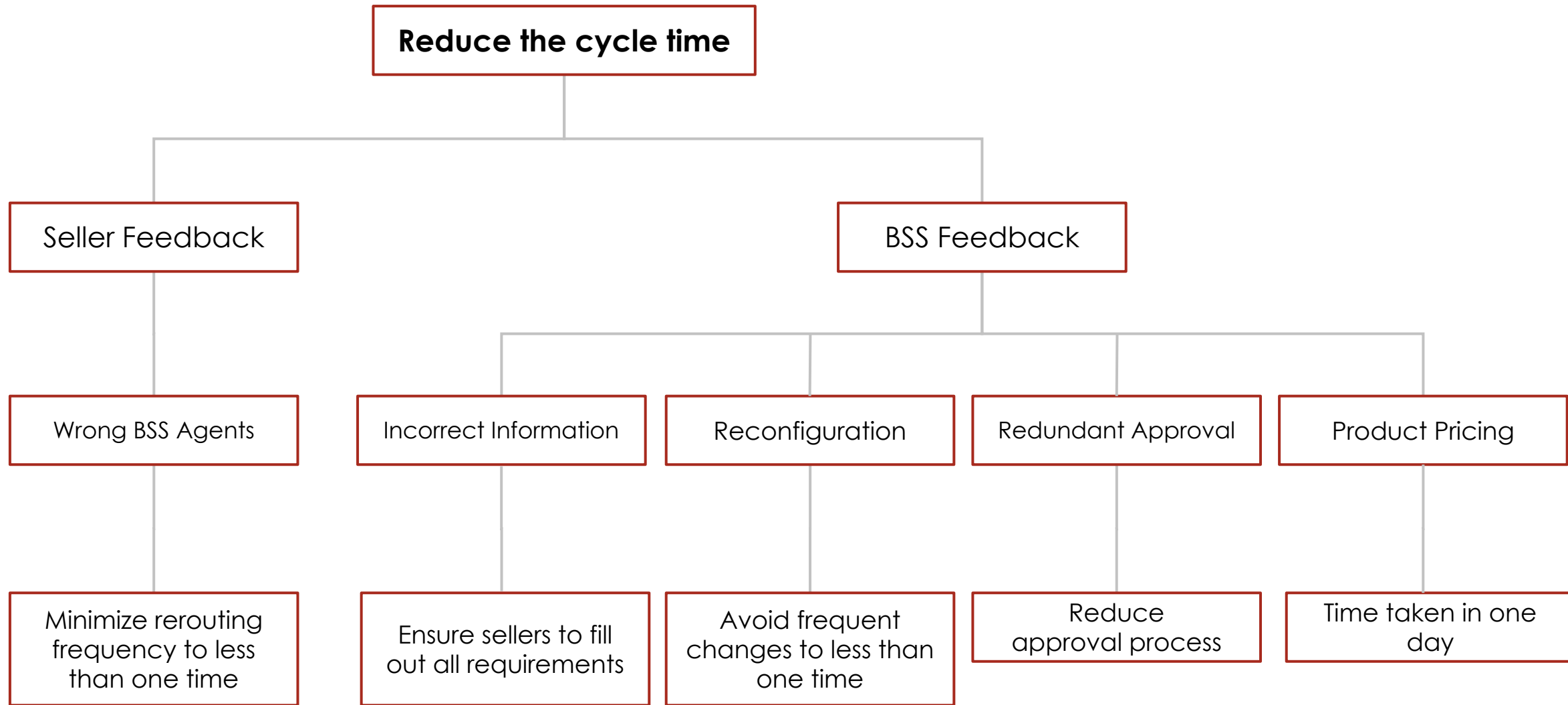
P O W E R	H I G H	Proposal Support Manager Brand Manager	Elliot Smith Grace Monroe Jeff Hugh
	L O W	Product Design team Pricing Group	Sellers Bid Support Specialist
		LOW	HIGH
		LEVEL OF INTEREST	

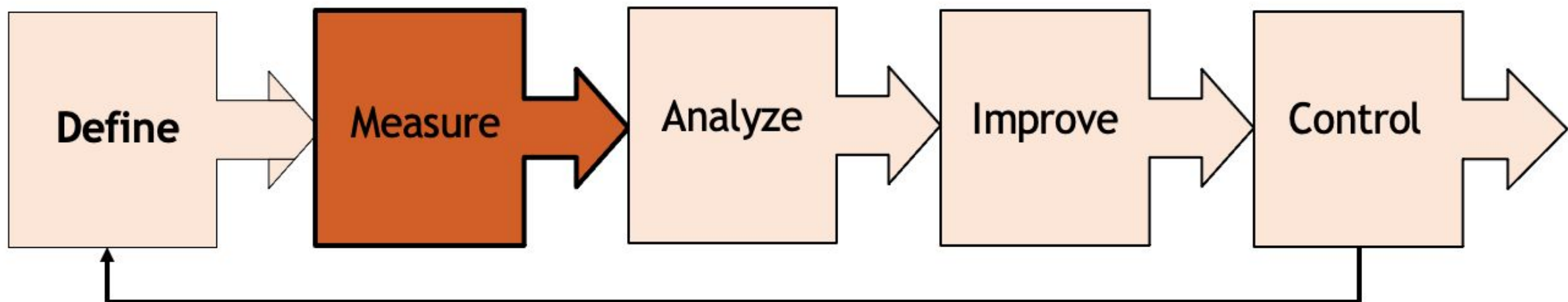


KANO MODEL



CTQ TREE





KPIs

KPI 1: Full Cycle Time	The whole proposal creation process time, the time elapsed from seller submit request to BSS until final proposal is ready to sent to client
PROCESS START	The time when the seller submit proposal request in the ERP system
PROCESS END	The time when the the final proposal is ready for client
AVERAGE DAYS FOR EACH STEP	31.6 DAYS

KPI 2: STS 1	Time spent on seller sending request in ERP
PROCESS START	The time when the seller submit proposal request in the ERP system
PROCESS END	The time when the request is record in the system
AVERAGE DAYS FOR EACH STEP	1.9 DAYS

KPI 3: ZQT2	Time spent until the request is routed to the right agent
PROCESS START	The time when the BSS receives and access the request
PROCESS END	The time when the the request is assigned to right agent
AVERAGE DAYS FOR EACH STEP	2 DAYS

KPI 4: ZQT3	Time elapsed till all required information is collected from seller
PROCESS START	When the agent check occurrence of missing information
PROCESS END	The time that required information is retrieved
AVERAGE DAYS FOR EACH STEP	3.1 DAYS

KPI 5: ZQT4	The time elapsed between the BSS review request with proposal support manager and the request is approved
PROCESS START	The time when the request is brought to the proposal support manager
PROCESS END	The time after the request is approved by the manager
AVERAGE DAYS FOR EACH STEP	3.6 DAYS

KPI 6: ZQT5	The time product design team spent to generate configuration that consists of technical specifications
PROCESS START	The time that the product design team receives approved request
PROCESS END	After product design team generates the configuration and send it to BSS for review
AVERAGE DAYS FOR EACH STEP	2 DAYS

KPI 7: ZQT6	The time elapsed between the BSS review configuration, reconfiguration process if changes made, and approval from the seller
PROCESS START	When BSS begin review configuration sent from product design team
PROCESS END	The time that the configuration is approved by seller
AVERAGE DAYS FOR EACH STEP	7.4 DAYS

KPI 8: ZQT7	The time frame between BSS send out request to the pricing team, pricing generated, and approved by brand manager
PROCESS START	The time when BSS send request to the pricing team
PROCESS END	When brand manager approves the pricing proposal
AVERAGE DAYS FOR EACH STEP	3.1 DAYS

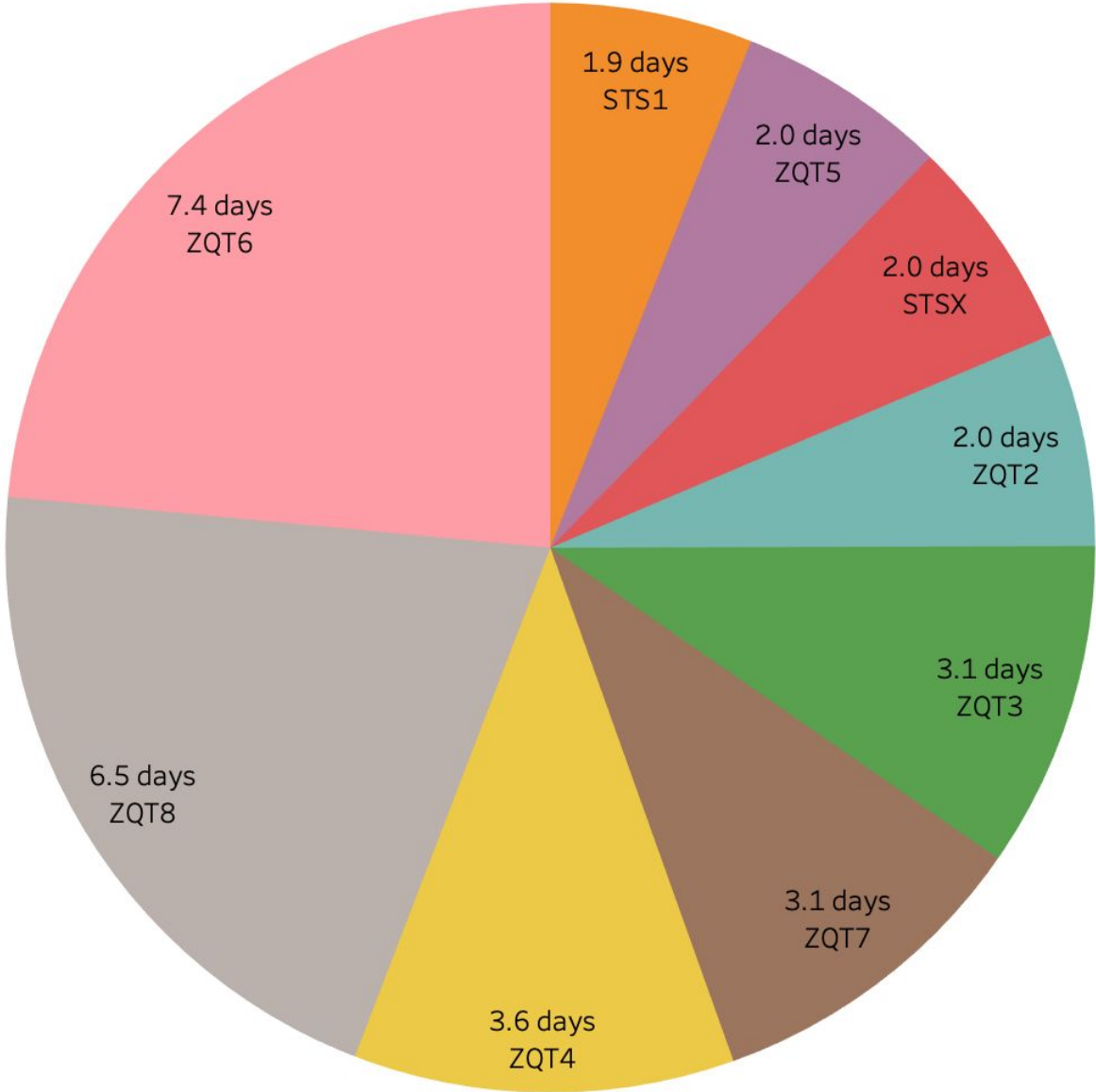
KPI 9: ZQT8	Time needed for BSS to prepare final document, product support manager to check proposal and terms and conditions, and submit to seller
PROCESS START	When BSS start to prepare final document for proposal after pricing is completed
PROCESS END	When the proposal support manager submit the final proposal to the seller
AVERAGE DAYS FOR EACH STEP	6.5 DAYS

KPI 10: STSX	The time requires for the seller to prepare the final proposal for the client
PROCESS START	The time when the seller receives final proposal
PROCESS END	The time when the the final proposal is ready for client
AVERAGE DAYS FOR EACH STEP	2 DAYS

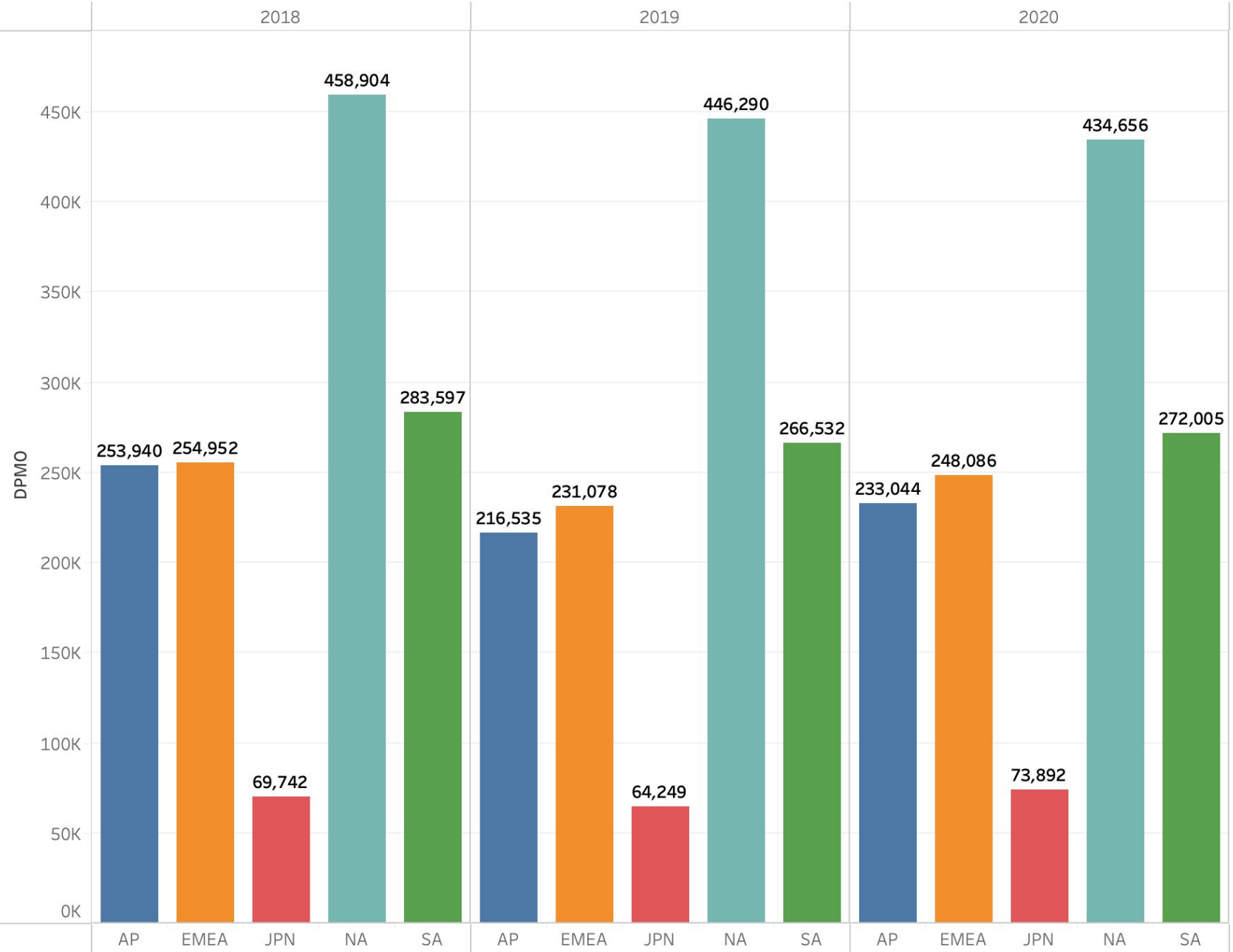
DEFECTS PER MILLION OPPORTUNITIES

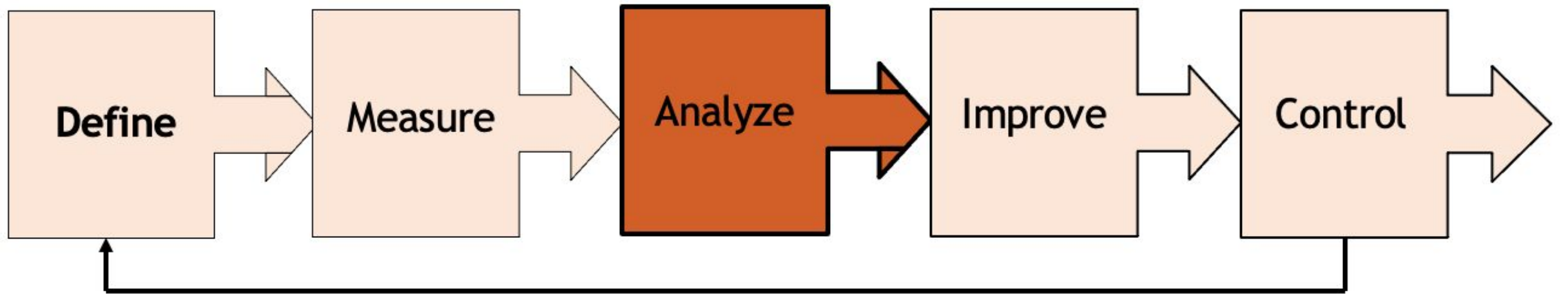
Average Cycle Time

TOTAL DEFECTS	21079
TOTAL OPPORTUNITIES	75000
DPMO	281053
Sigma Level	2.08

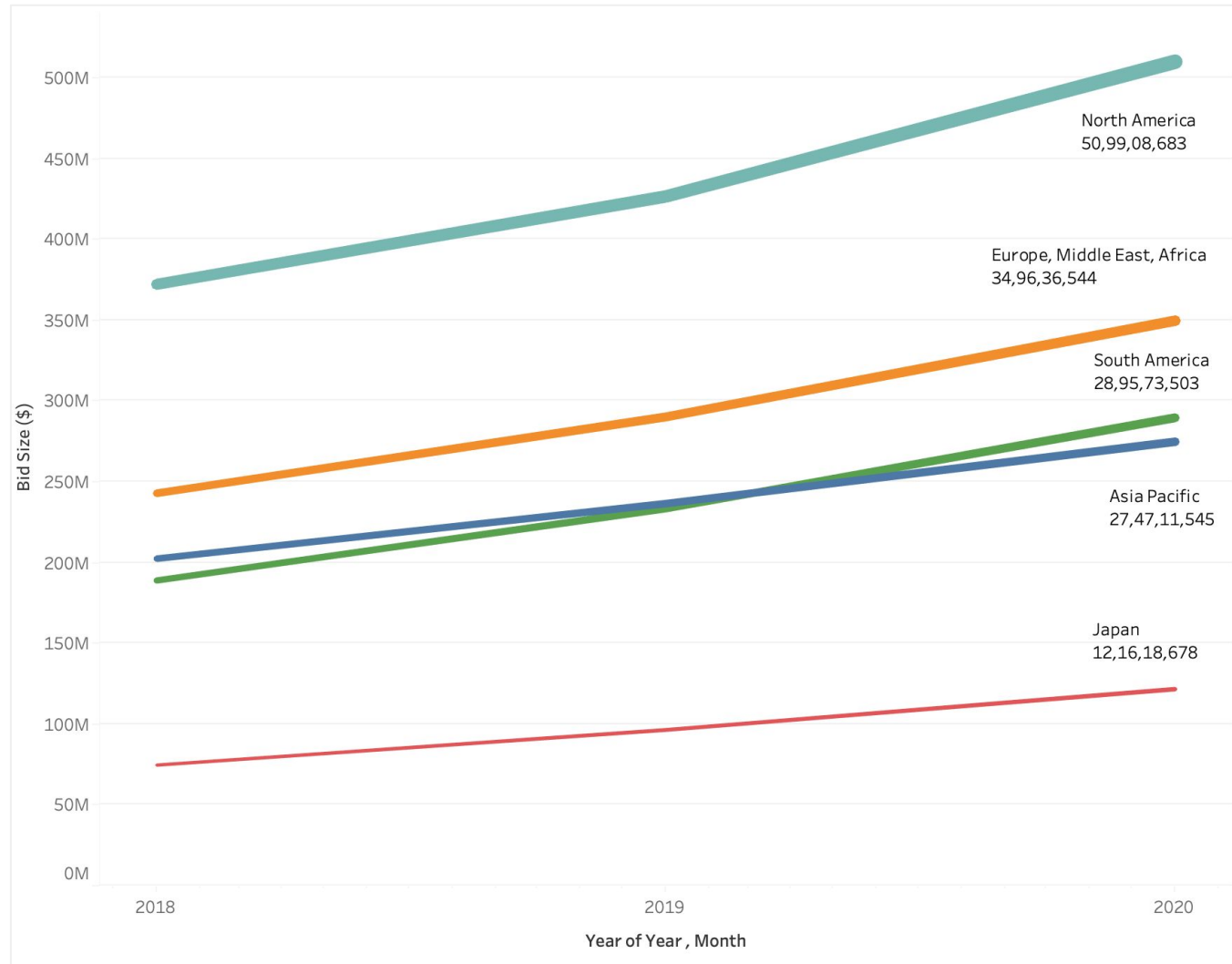


DPMO by Year & Geo

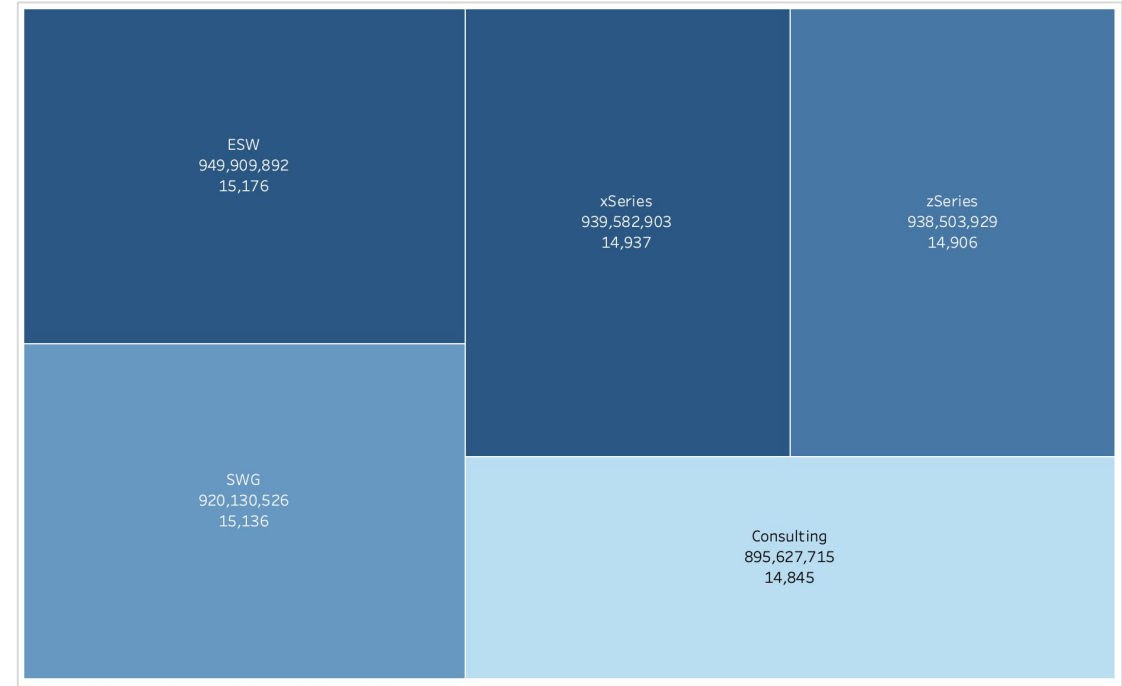


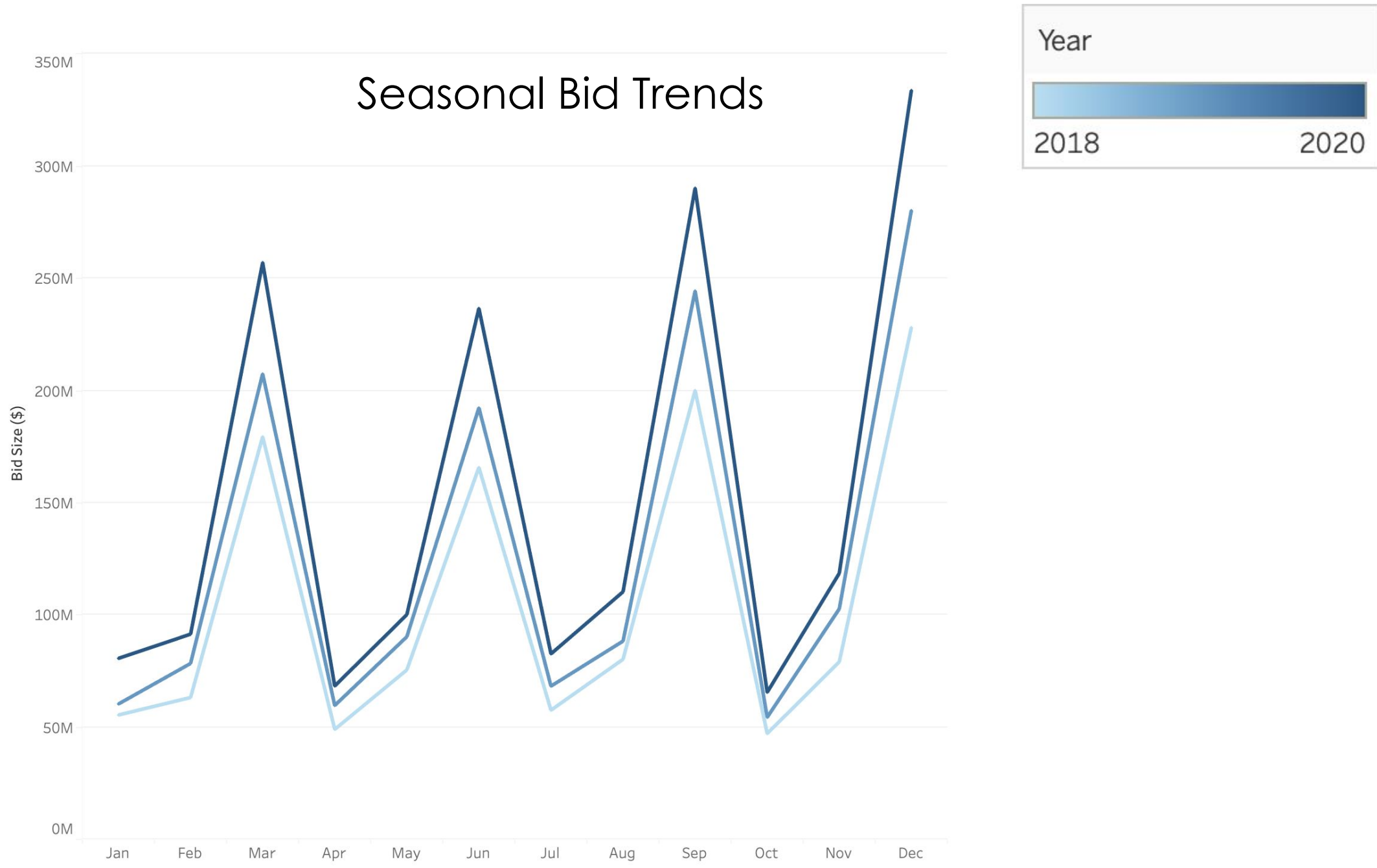


Sales by Geo and Year

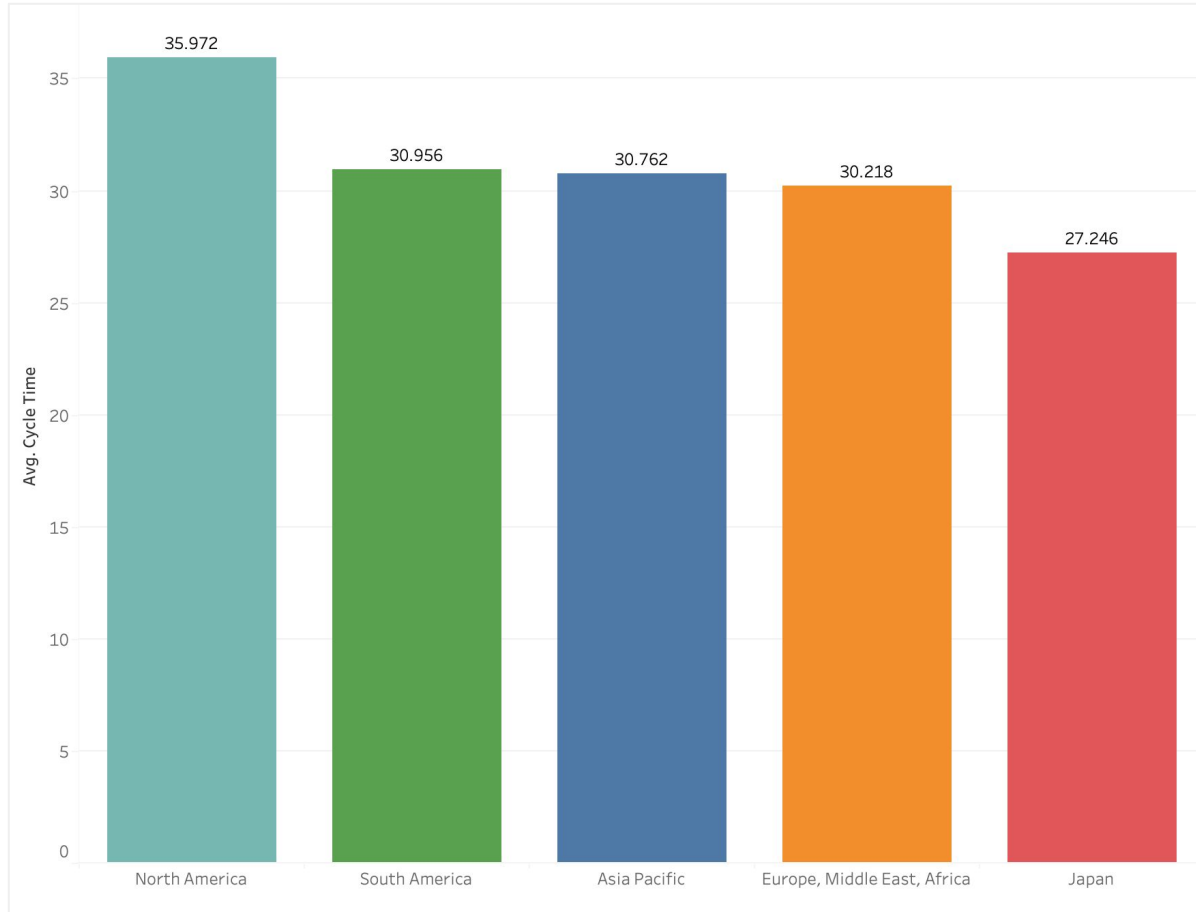


Sales by Brand

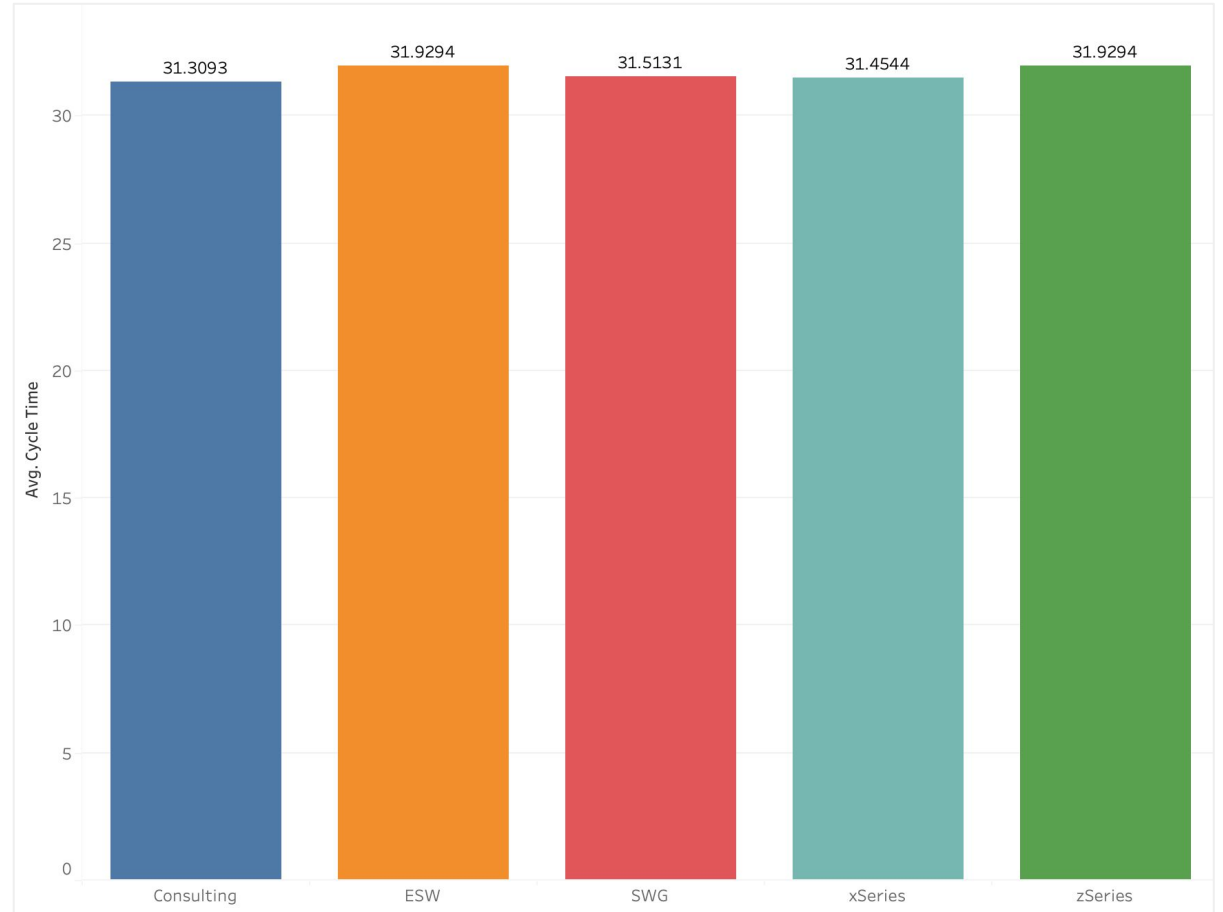


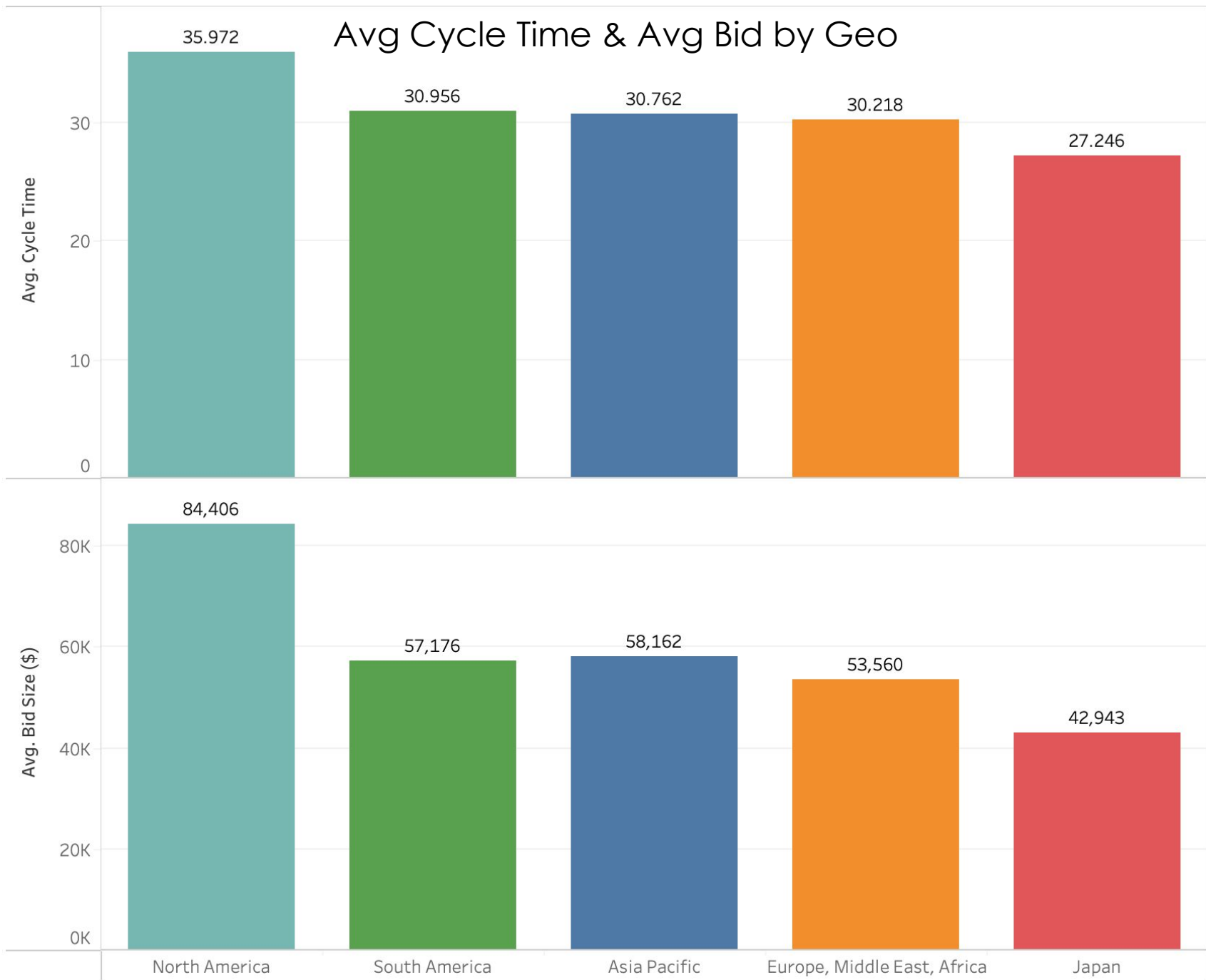


Avg Cycle Time by Geo

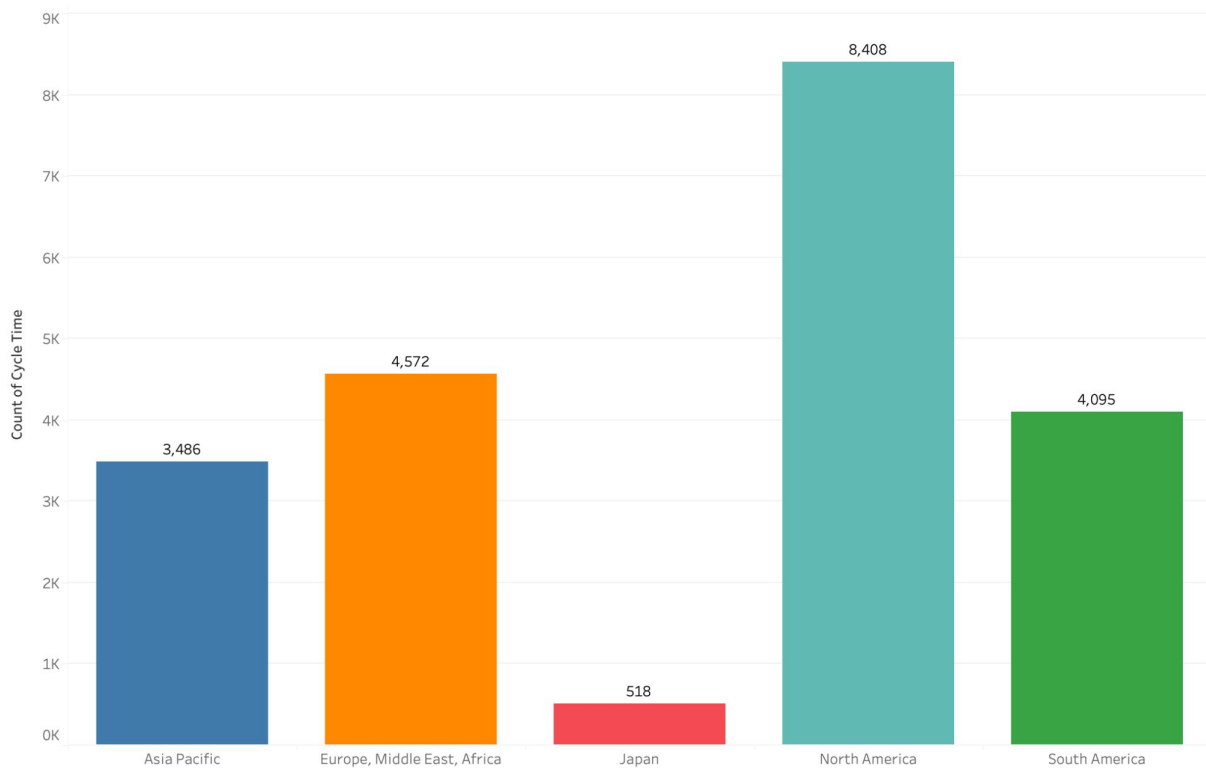


Avg Cycle Time by Brand

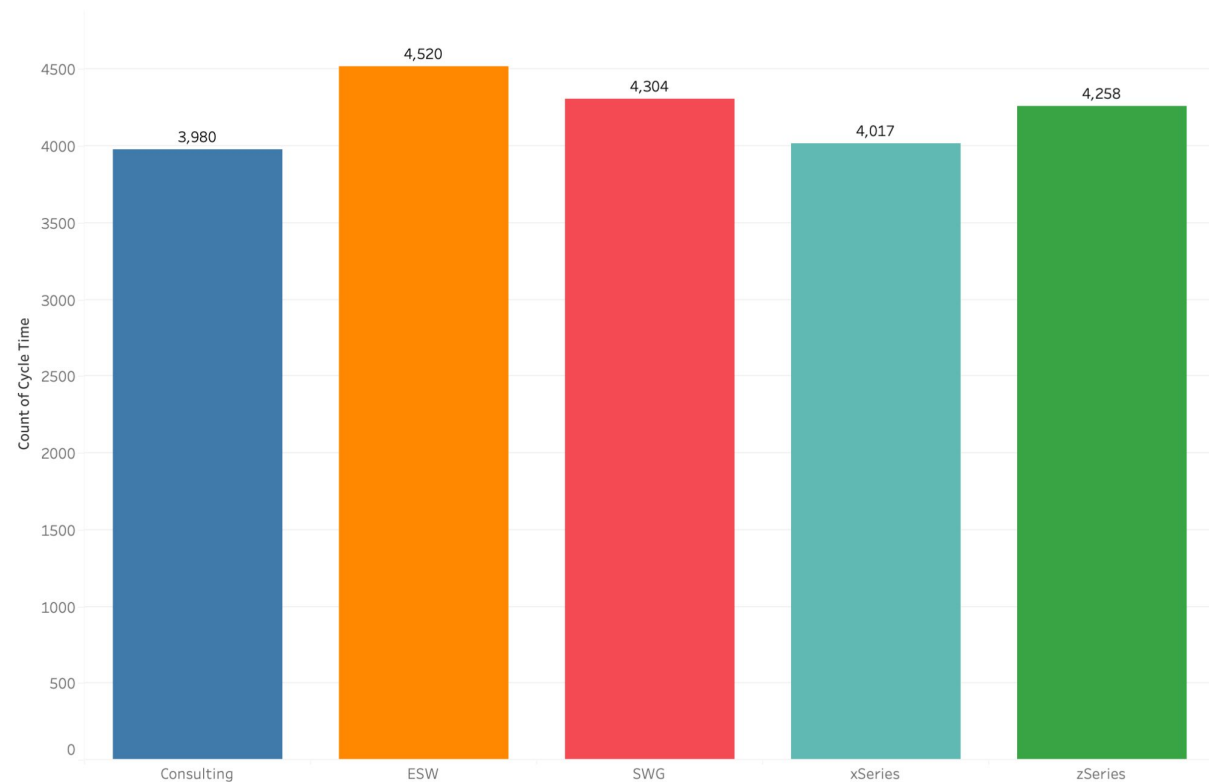




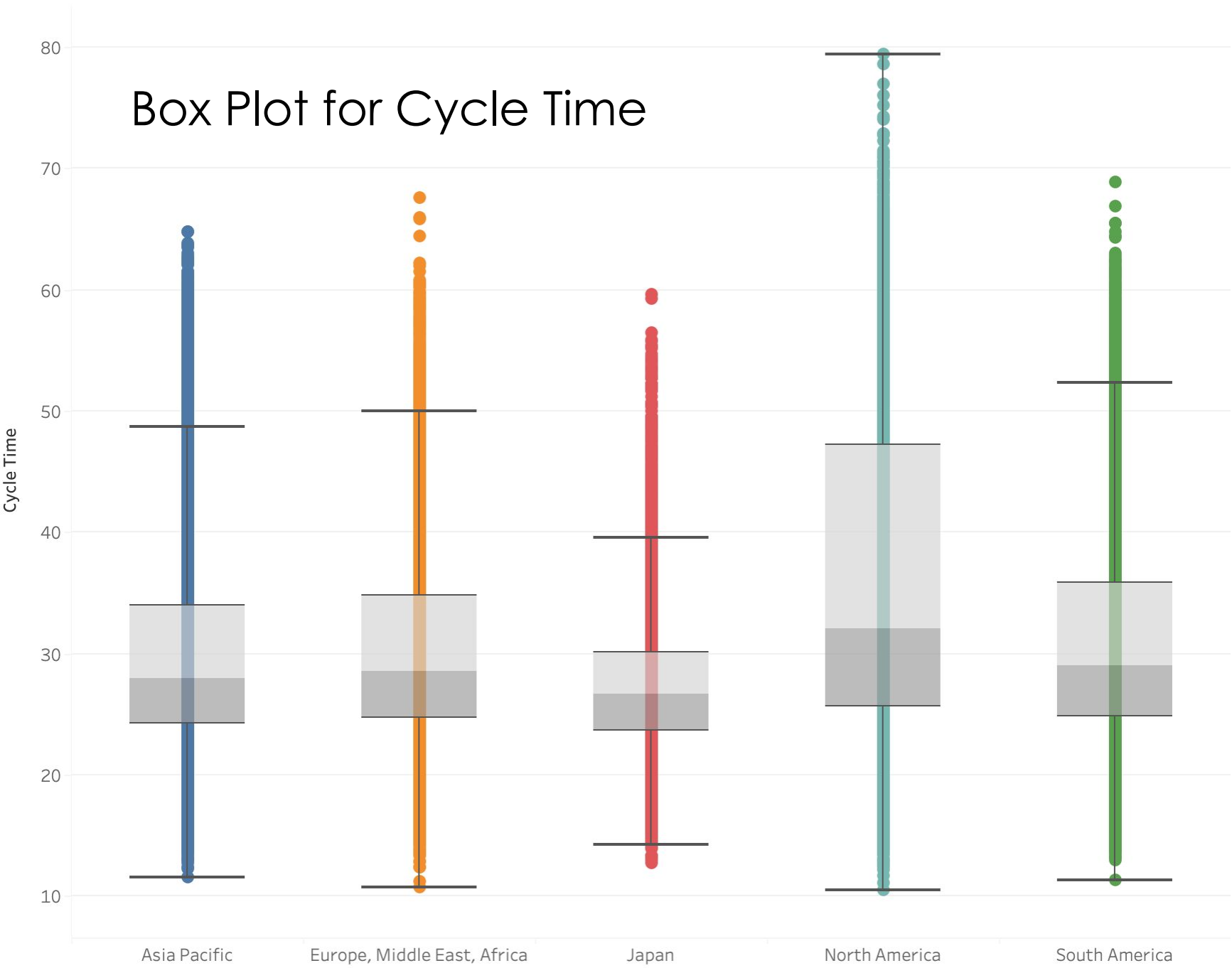
Defects by Geo

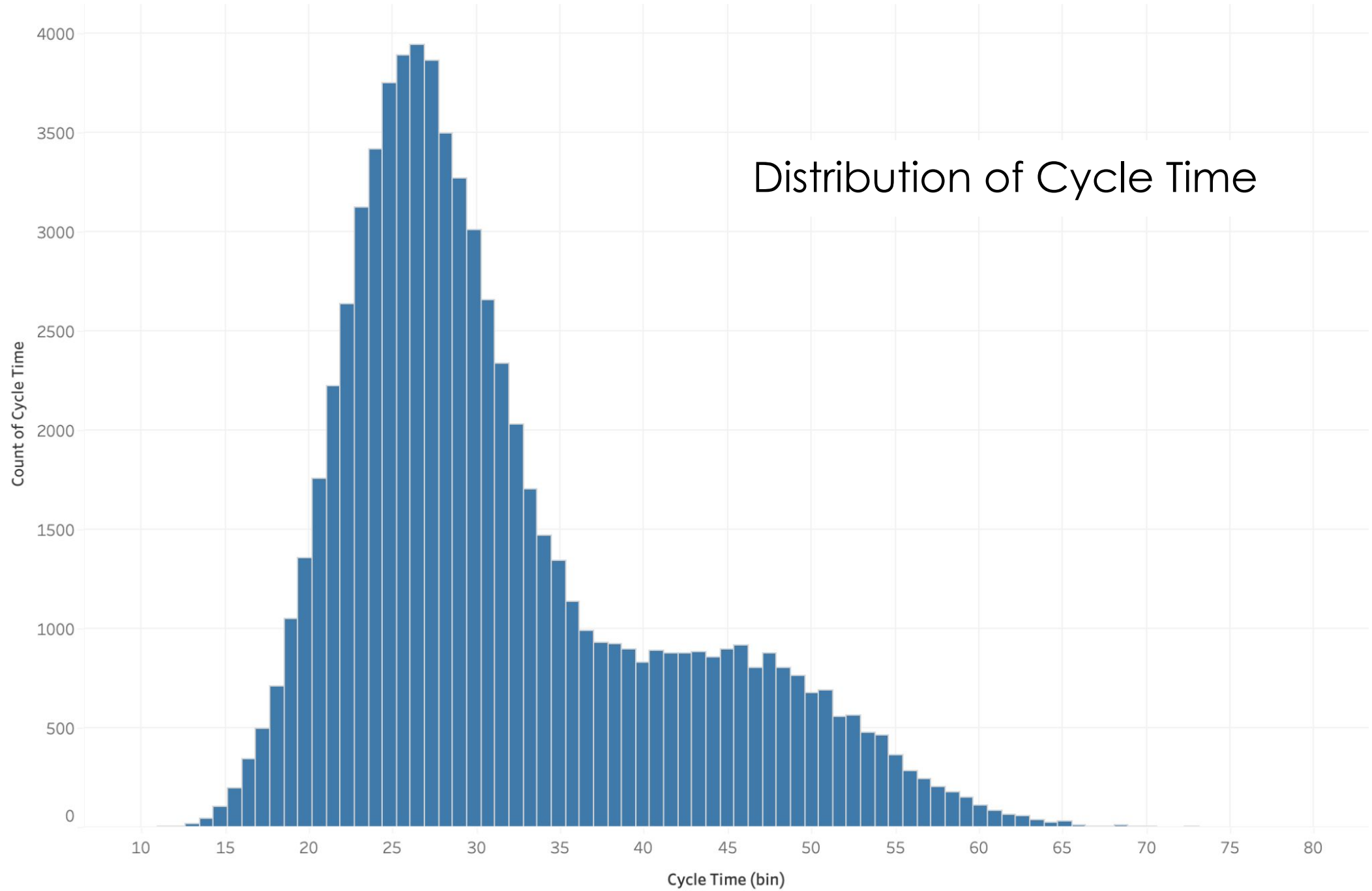


Defects by Brand

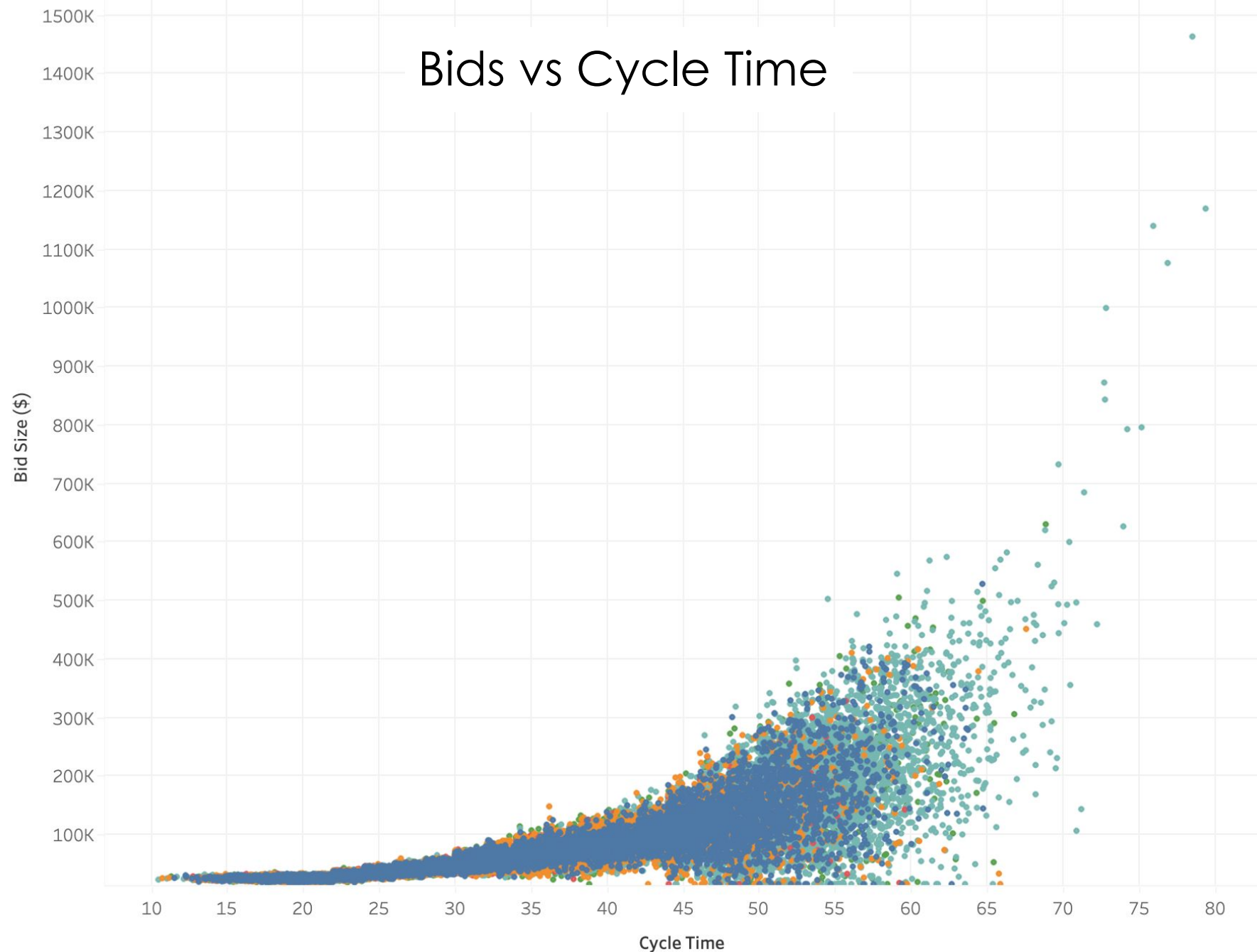


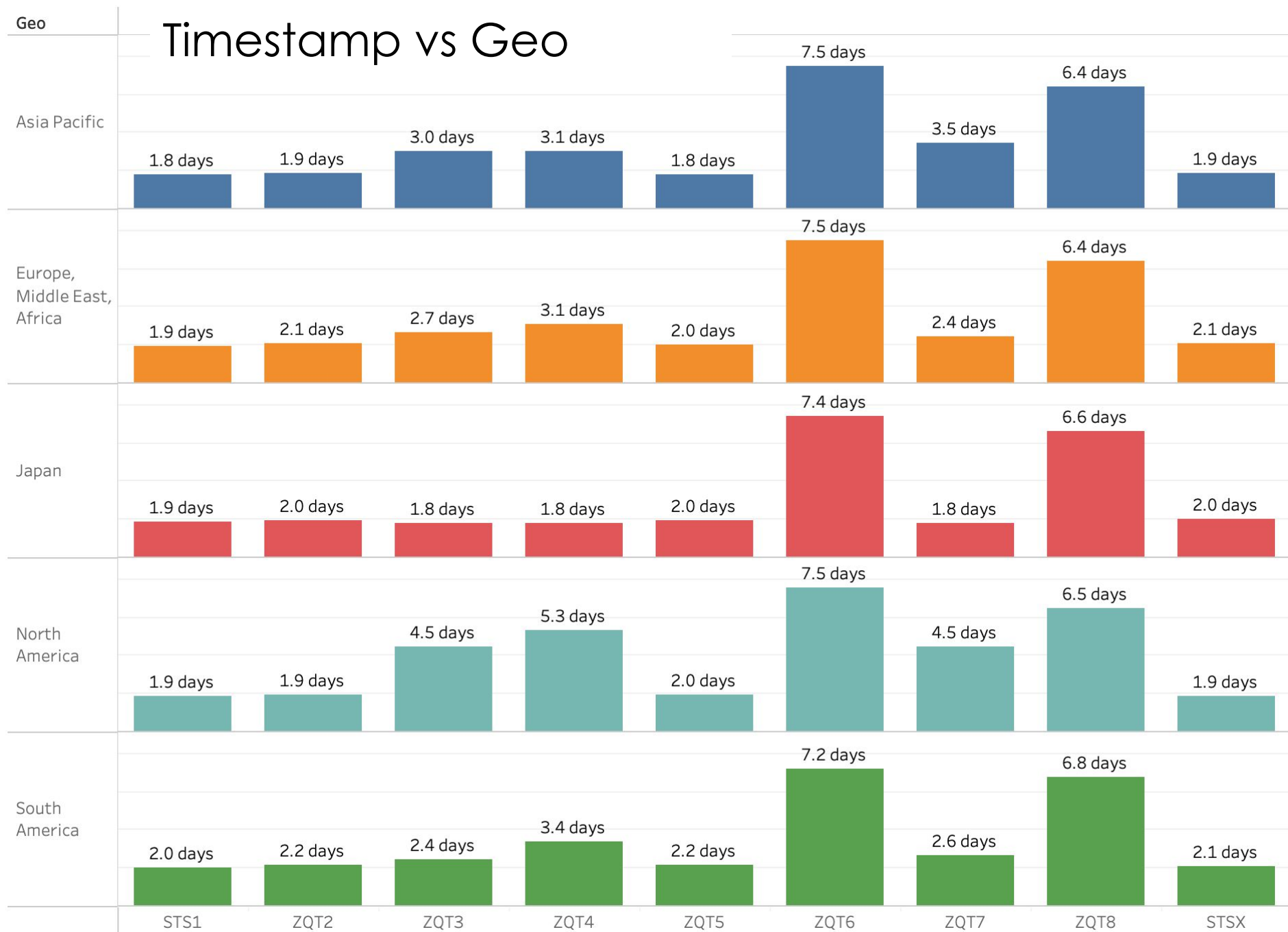
Box Plot for Cycle Time

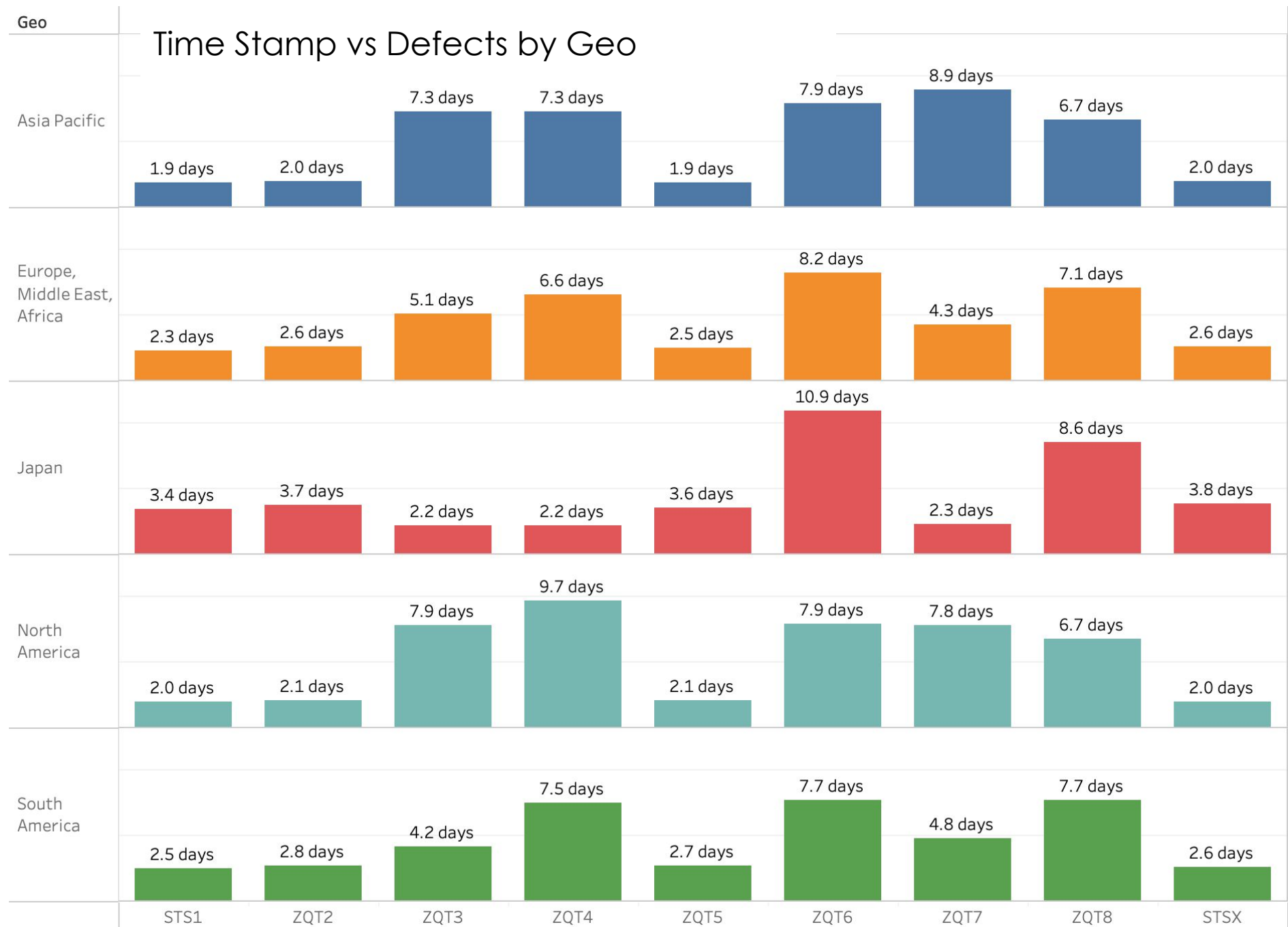


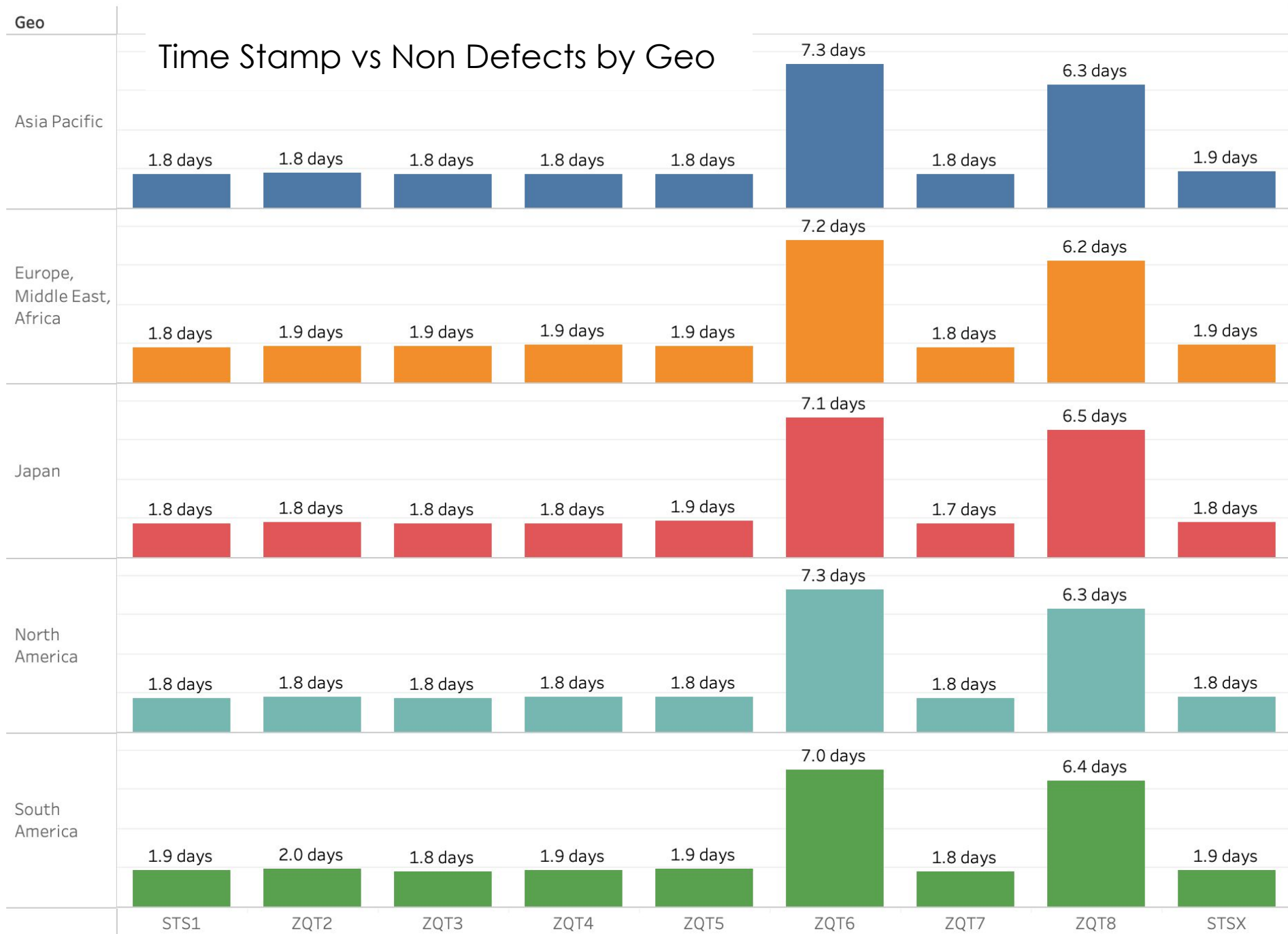


Bids vs Cycle Time

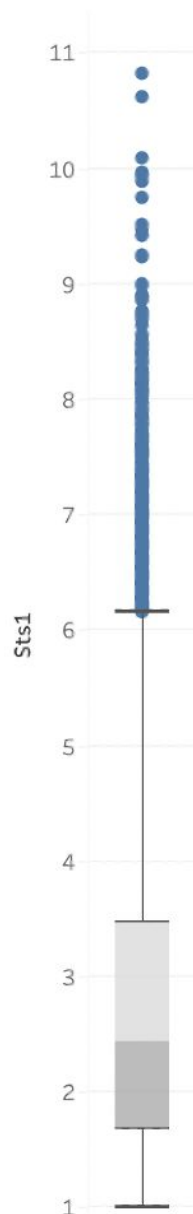




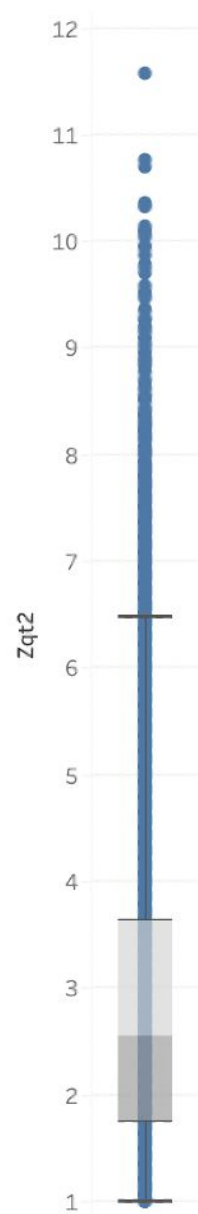




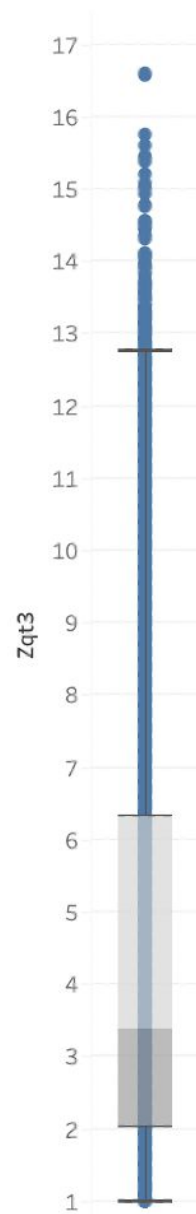
Box - STS1



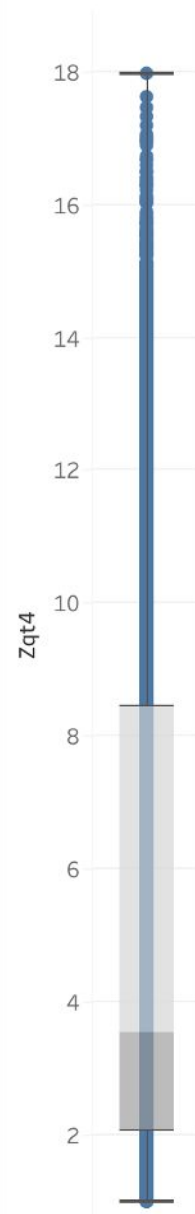
Box - ZQT2



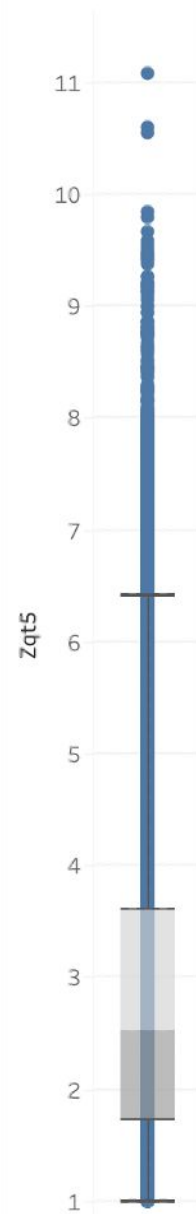
Box - ZQT3



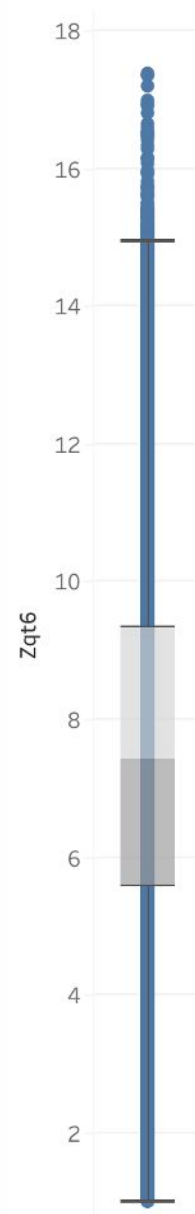
Box - ZQT4



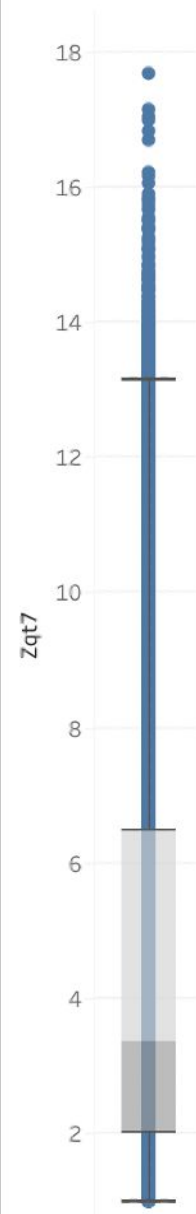
Box - ZQT5



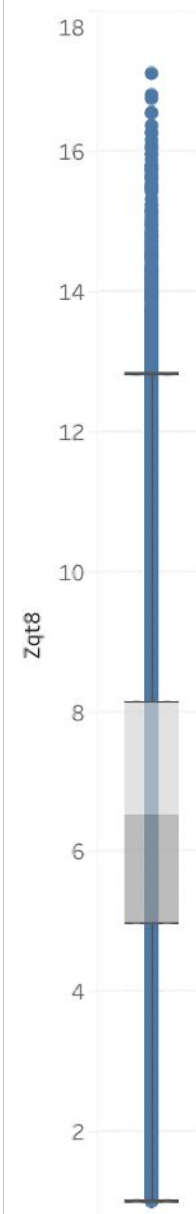
Box - ZQT6



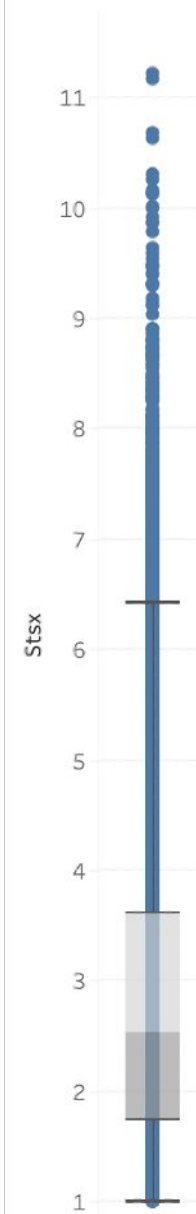
Box - ZQT7



Box - ZQT8

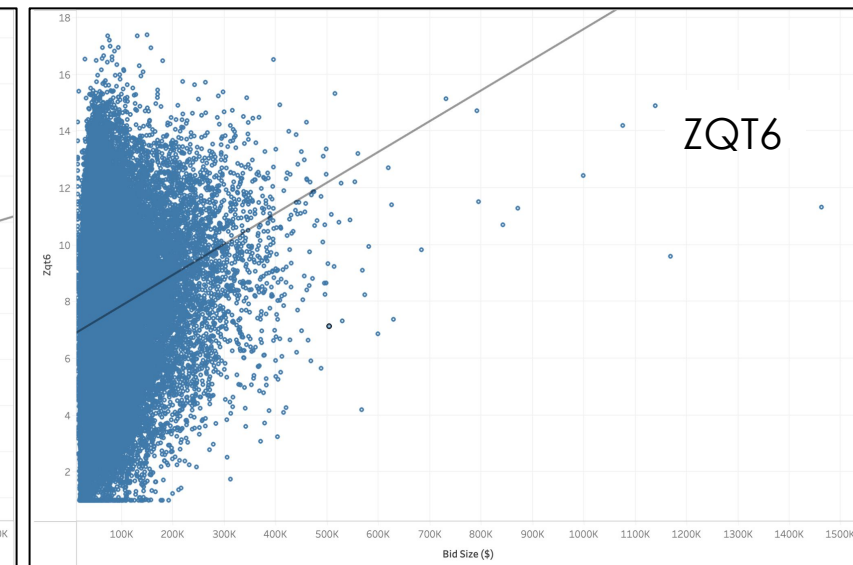
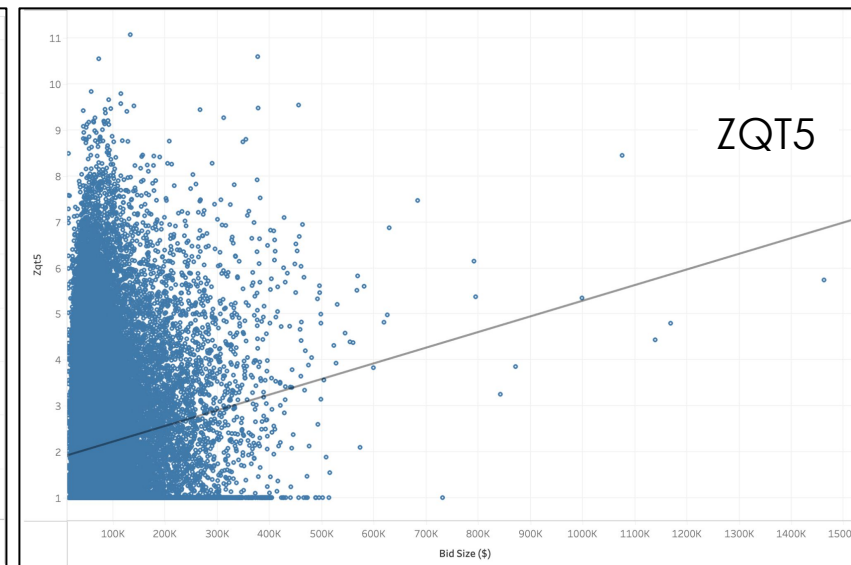
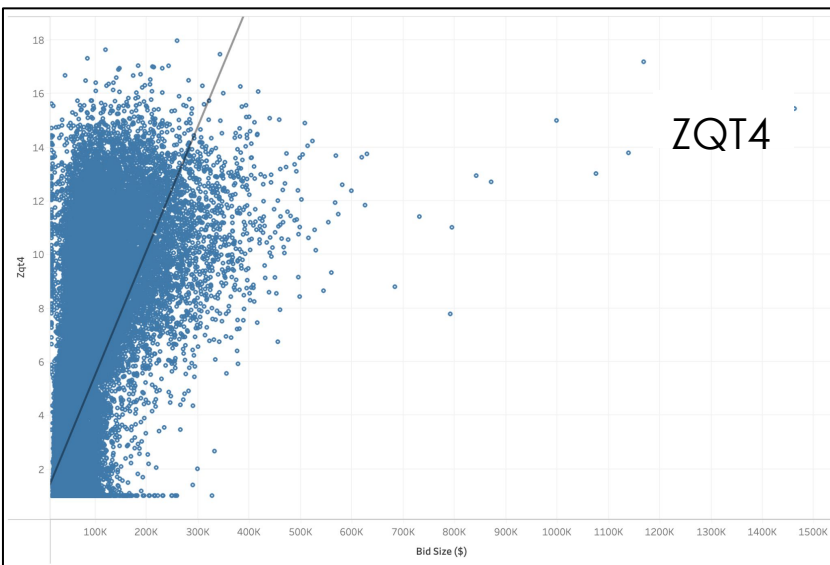
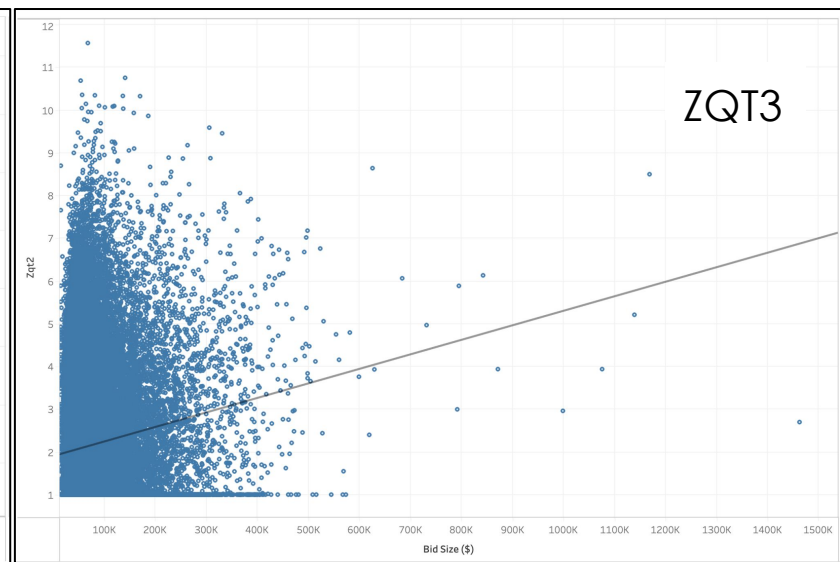
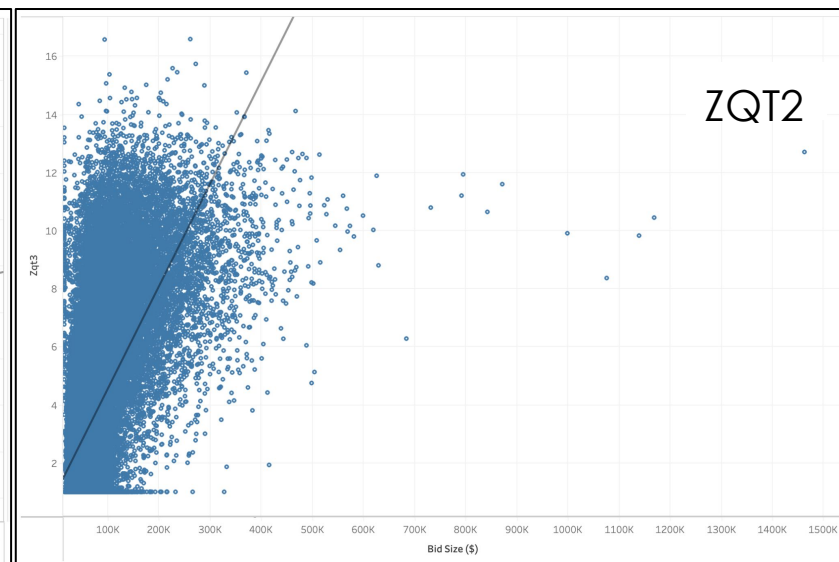
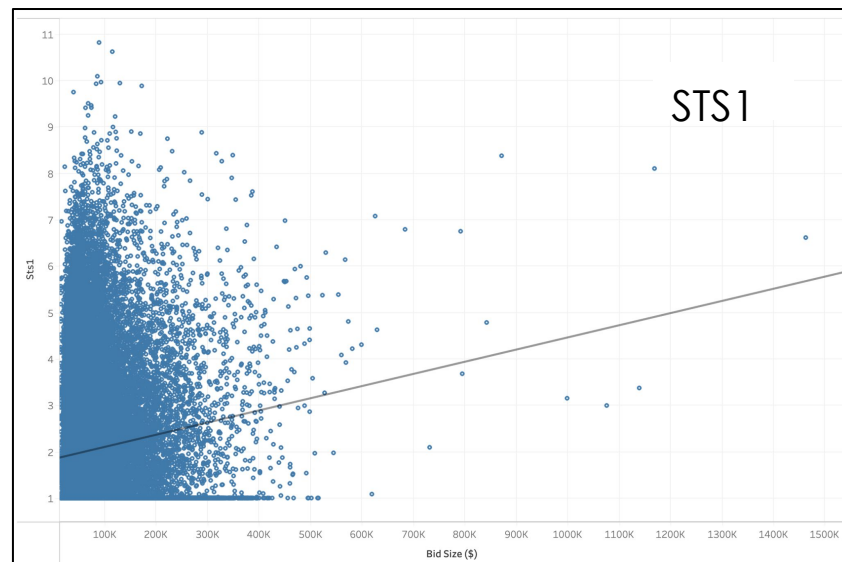


Box - STSX

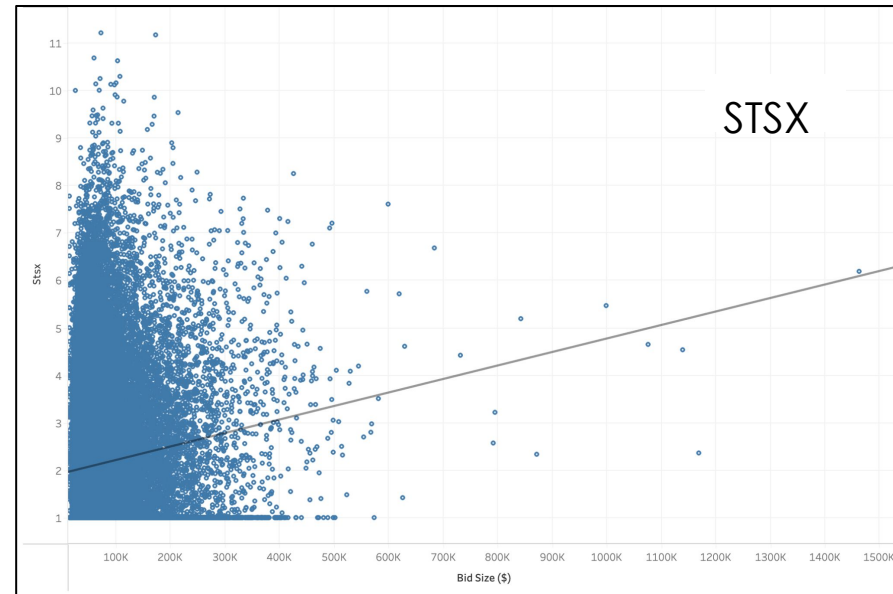
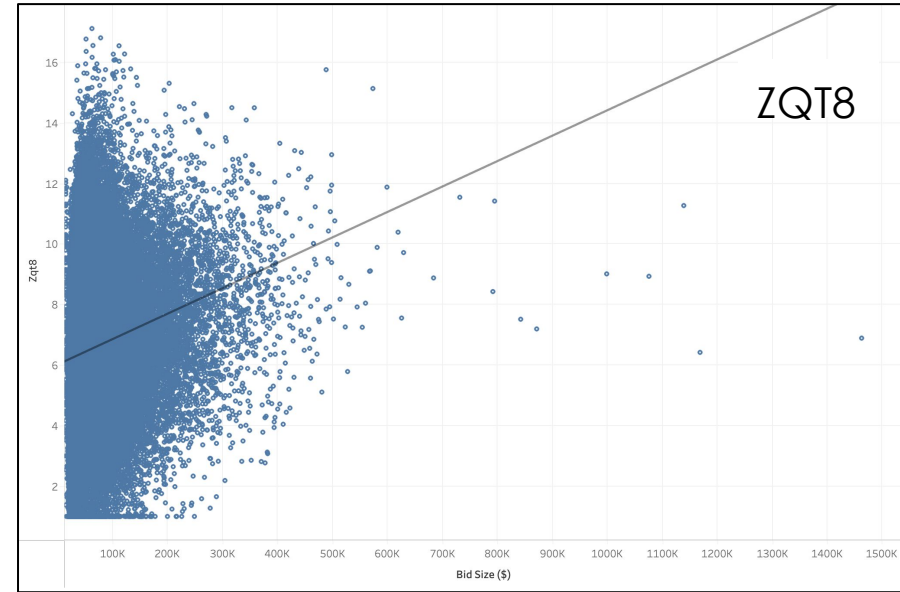
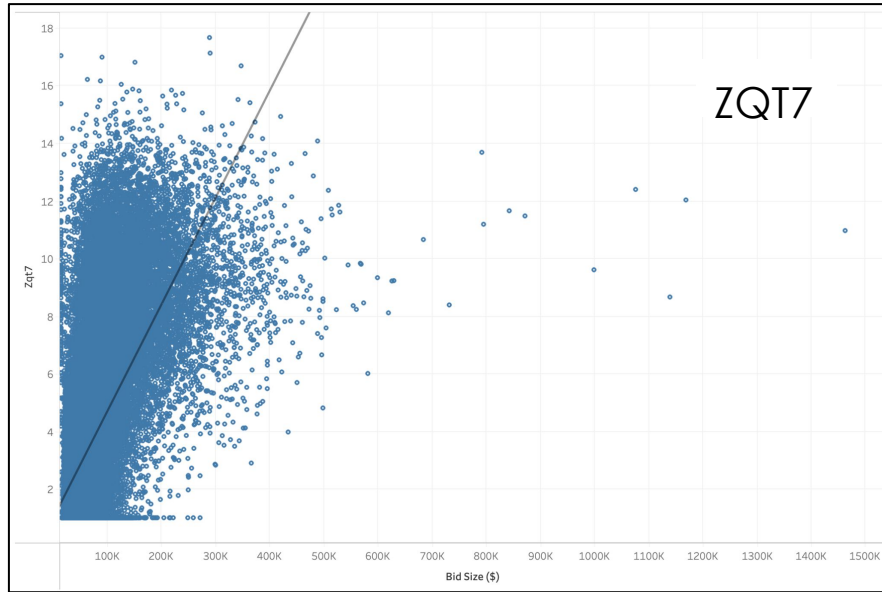


Timestamp
Analysis

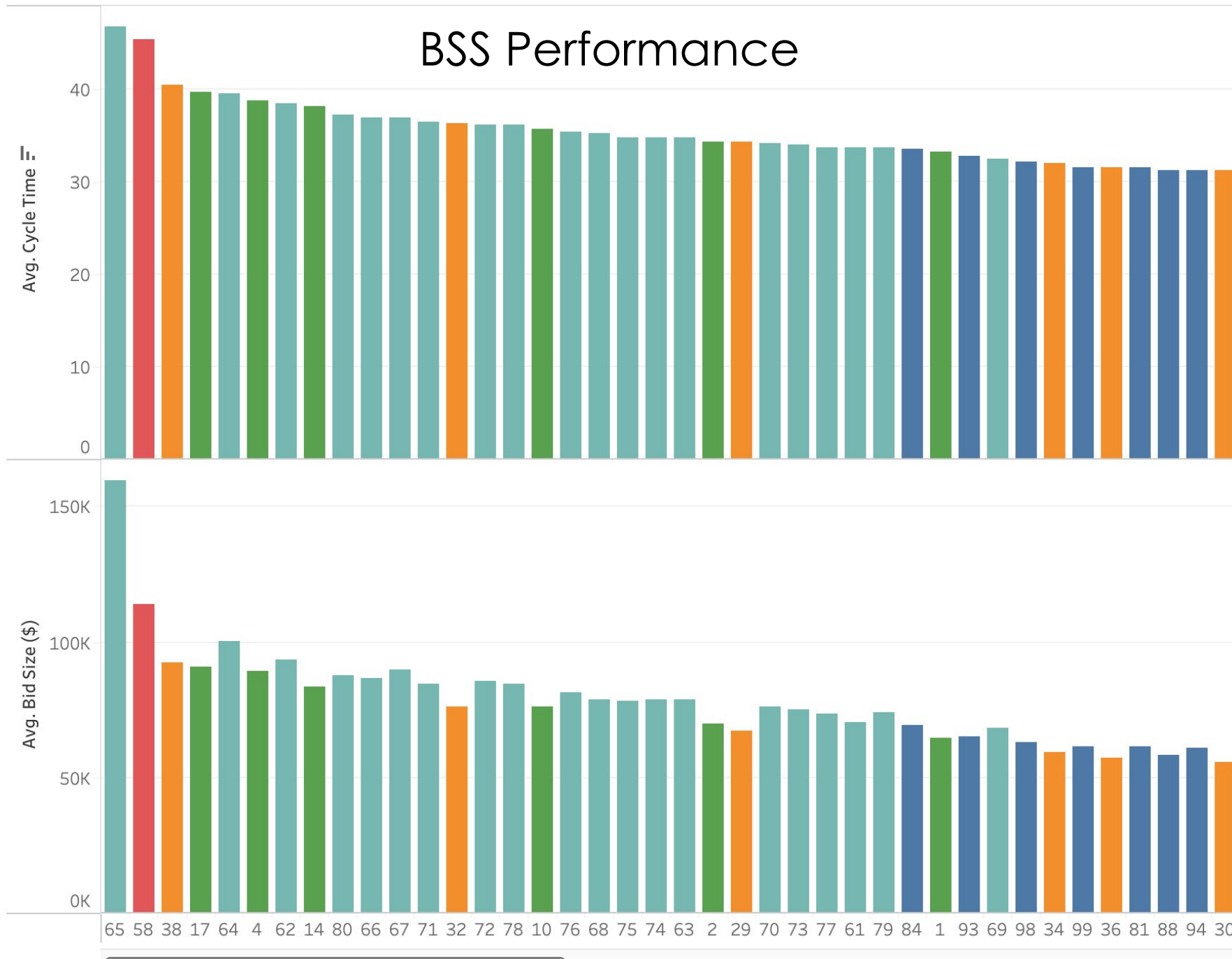
Correlation of Cycle time and Bid size for each time stamp



Correlation of Cycle time and Bid size for each time stamp



BSS Performance





VALUE ADDED AND NON-VALUE ADDED

Proposal Creation Process Steps:

Value Added

IDENTIFY THE SALES OPPORTUNITY

SUBMITS THE REQUEST IN ERP SYSTEM

GENERATING TECHNICAL SPECIFICATION

PRICING TEAM GENERATES PRICING
SOLUTION BASED ON CONFIG

PREPARATION OF FINAL DOCUMENTS

Non Value Added but Needed

APPROVAL OF PROPOSAL

CONFIG APPROVAL BY SELLER

APPROVAL OF PRICING PROPOSAL

REVIEW OF FINAL DOCUMENTATION AND T&C

Non Value Added and Not Needed

BSS ACCESSES COMPLEXITY OF REQUEST

RETRIEVAL OF MISSING INFORMATION

BSS REVIEW CONFIG FOR COMPLETION

RECONFIGURATION OF PROPOSAL

5 WHYS

Why were changes in team dynamics, roles, or responsibilities not adequately considered?

There may not have been a formal process for regular review and adjustment of BSS assignments because of changing time and situation.

Why was the process not completely reviewed or updated?

Because the changes of team dynamics, roles, or responsibilities were not adequately considered during the initial setup of the process.

Why are wrong BSS agents assigned?

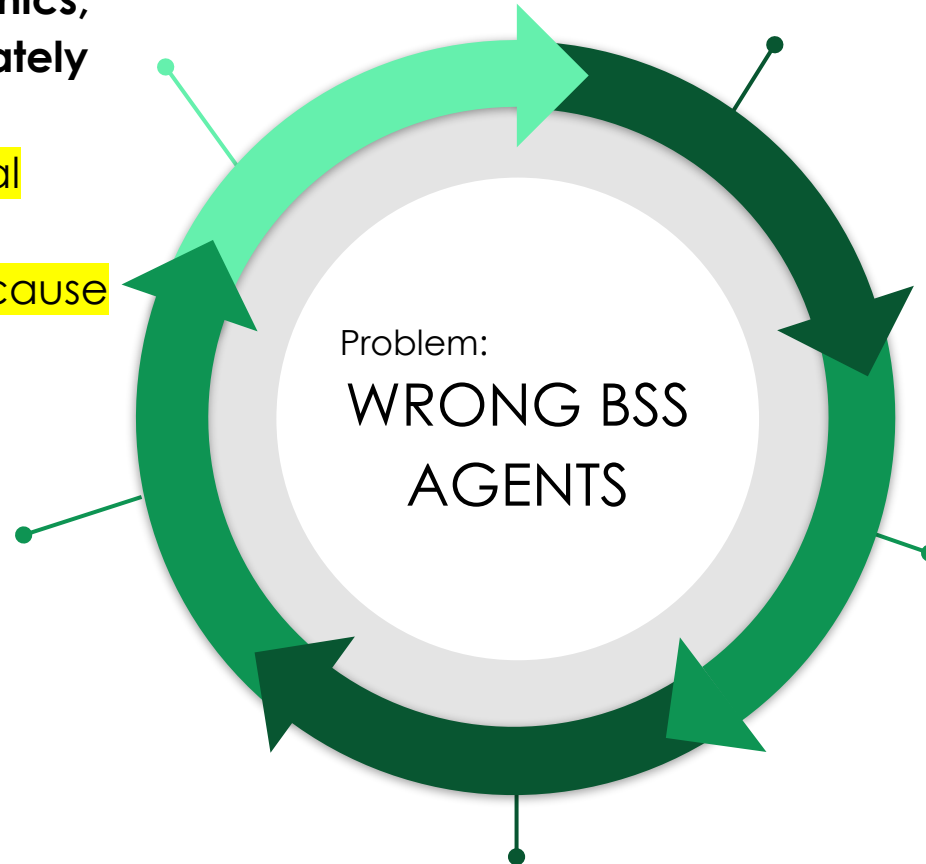
Because there may not be a clear request submission process regarding which BSS agent is appropriate for each request.

Why is there no clear request submission in process?

Because there may not be a standardized system or rules for Sellers to make sure the correct BSS agent.

Why is there no standardized system of rules for Sellers?

Because the process for assigning BSS agents may not have been completely reviewed or updated to meet their needs.



5 WHYS

Why may there be insufficient training or communication and why no system to control the information?

The training program for Sellers may not have been comprehensive or may not have emphasized the importance of complete information and also the initial system setup may not think of avoiding missing values.

Why may the sellers not notice the importance of complete information?

There may not be enough training or communication for sellers, or no system to control the information.

Why is information often missing from sellers?

Because Sellers may be in a hurry to submit requests or may not have clear rules on what information must be provided.

Why are Sellers in a hurry to submit requests or do not have clear rules to follow?

Because they may have some pressure to meet deadlines or do not notice the importance of providing complete information.

Why is there pressure to meet deadlines?

Because the market is competitive, it is required to have faster work.



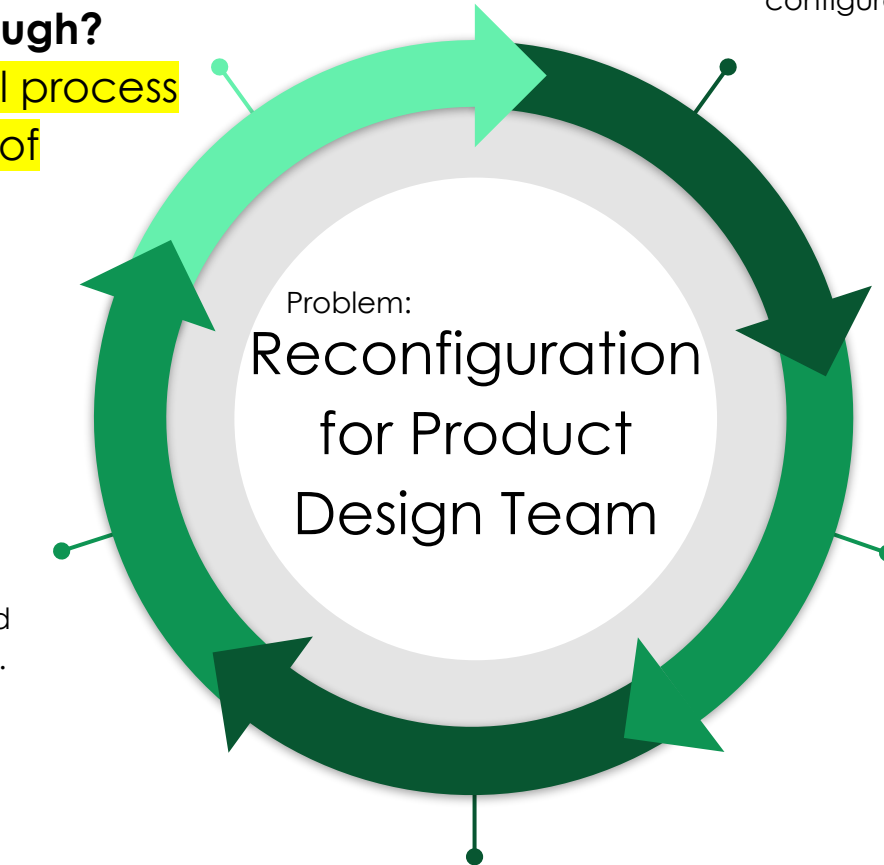
5 WHYS

Why are changes of employees, work and responsibilities not considered enough?

There may not have been a formal process for regular review and adjustment of communication.

Why is the process not completely reviewed or updated?

There may be changes of employees, work and responsibilities and may not have been considered enough during the beginning setup of the process.



Why is reconfiguration needed for the Product Design team?

Because there may be ineffective communication between the BSS and the Product Design team, leading to misunderstandings or incomplete configurations.

Why is there ineffective communication?

It may be because there may not be standardized procedures or tools for communicating reconfiguration requests.

Why are there no standardized procedures or tools for communication?

The process for handling reconfiguration requests may not have been completely reviewed or updated to ensure clear and efficient communication.

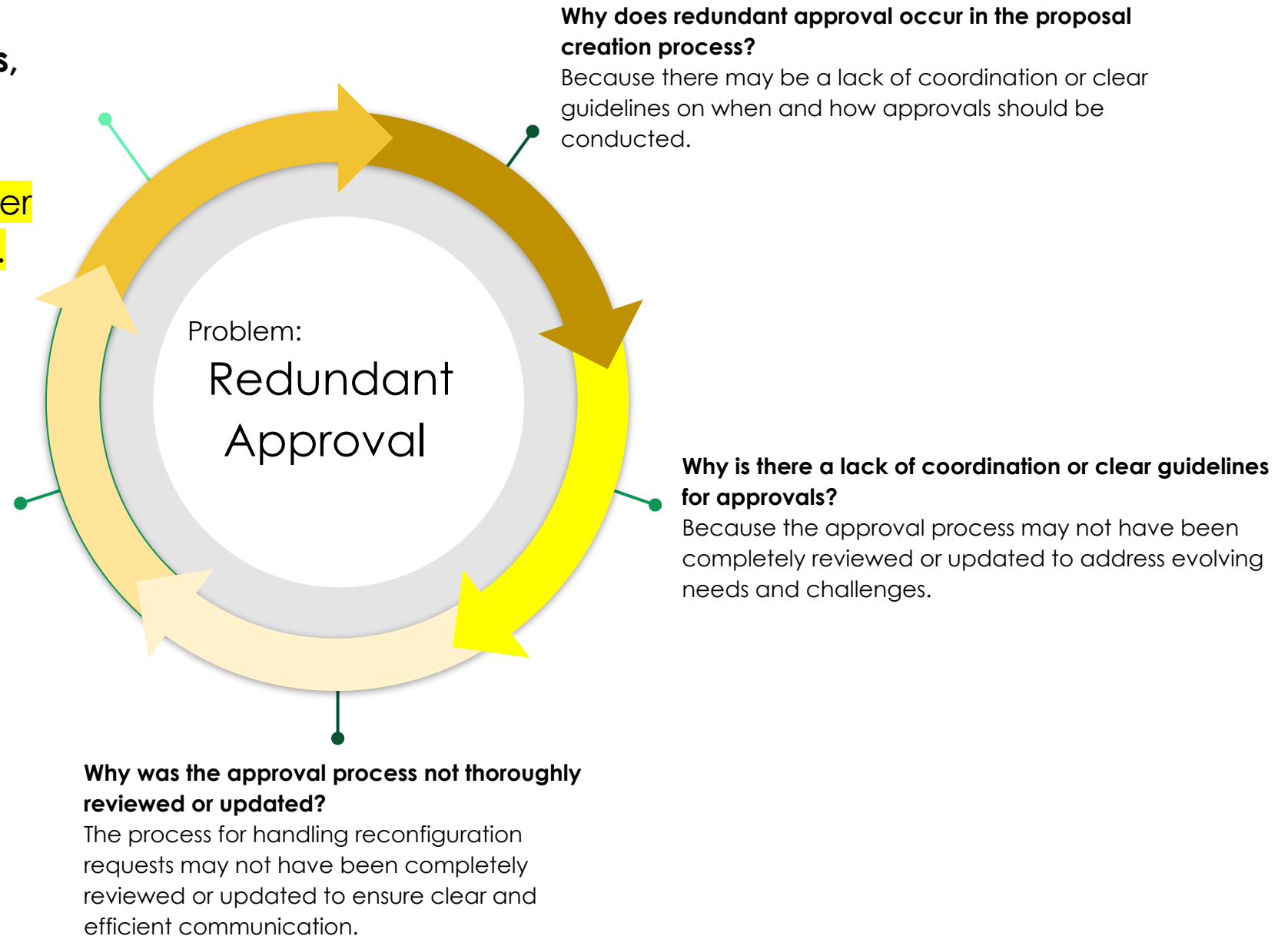
5 WHYS

Why are changes of employees, work and responsibilities not considered enough?

The initial setup may not consider the time spending on approval.

Why is the process not completely reviewed or updated?

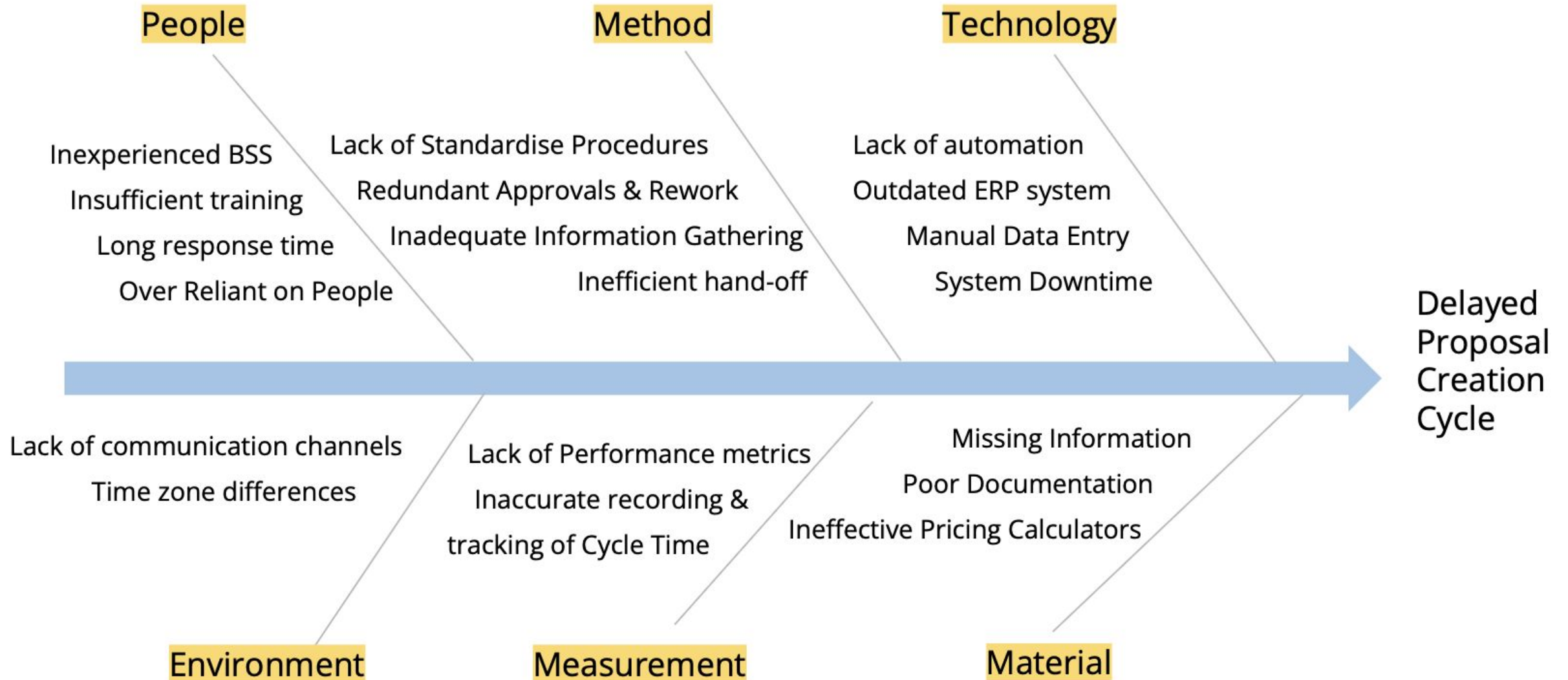
There may be changes of employees, work and responsibilities and may not have been considered enough during the beginning setup of the process.

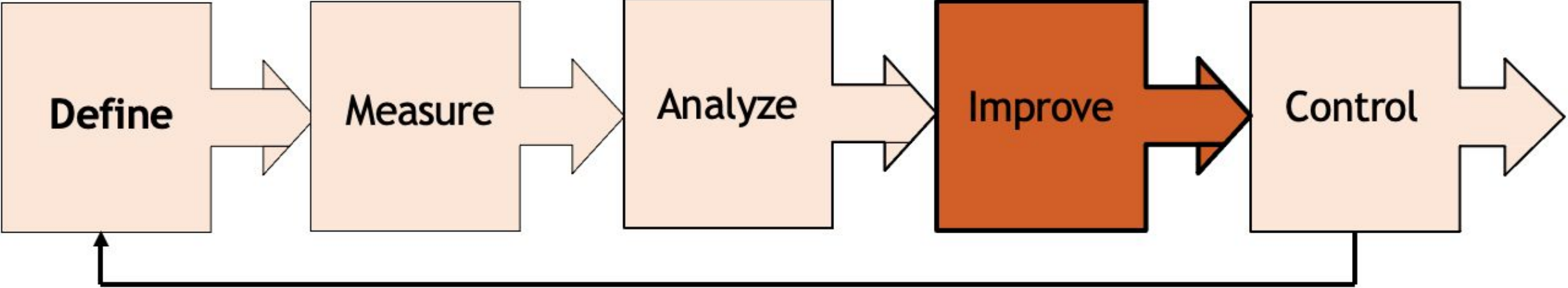


FMEA

Process Step	Potential Failure Mode	Potential Failure Effects	SEV	Potential Causes	OCC	Current Controls	DET	RPN
ZQT2	Wrong BSS Agents	Proposal delay	4	Lack of BSS training	7	Checking request	4	112
ZQT3	Missing Information	Proposal delay	5	Hurry to submit or incomplete requirement	7	Checklist for submission	3	106
ZQT6	Reconfiguration	Proposal delay	6	In effective communication	8	Configuration reviewed for submission	4	182
Approval (ZQT4,6,7)	Redundant approval	Proposal delay	6	Bad coordination	8	Peer review	5	240

FISHBONE ANALYSIS





Establish Required Fields in Proposal Request (Poka-Yoke)

Current Proposal:

Implement a 'mistake-proofing' approach by establishing mandatory fields in the proposal request form to prevent incomplete submissions.

Enhanced Improvements:

Conditional Logic in Forms:

Use conditional logic in electronic forms so that the information required can dynamically adjust based on previous answers, ensuring all necessary details are gathered.

Real-time Validation:

Implement real-time validation checks that prompt users for corrections before submission, reducing the need for follow-up.

Training and User Guides:

Provide comprehensive training and user guides to ensure that Sellers understand the importance of each field and how to complete them correctly.

Match Bid Complexity with BSS Experience Level to Avoid Rerouting

Current Proposal:

Match each proposal request with a Bid Support Specialist (BSS) based on the complexity of the bid and the experience level of the BSS.

Enhanced Improvements:

Dynamic Skills Matrix:

Develop a dynamic skills matrix that categorizes BSS agents by their areas of expertise and levels of experience. Use this matrix for real-time assignment of bids to the most suitable BSS.

Automated Matching Algorithm:

Implement an algorithm within the ERP system that automatically assigns bids to BSS agents based on the matrix, reducing the need for manual intervention.

Feedback Loop:

Create a mechanism for BSS agents to provide feedback on the matching process to continuously refine the matching criteria and algorithm.

Establish SOP for Proposal Submission

Current Proposal:

Create a Standard Operating Procedure (SOP) for proposal submission to increase accuracy and reduce the need for reconfiguration.

Enhanced Improvements:

Visual SOPs:

Incorporate visual elements such as flowcharts and diagrams in the SOP to enhance understanding and retention.

SOP Accessibility:

Make the SOP readily accessible, for instance, through mobile devices or integrated into the ERP system, so it can be easily referenced.

Regular SOP Reviews:

Set up a schedule for regular reviews and updates to the SOP to ensure it stays relevant and effective as processes and technologies evolve.

Implement Robotic Process Automation (RPA) for Pricing

Current Proposal:

Use RPA to automate the generation of pricing proposal for the Pricing Team

Enhanced Improvements:

Customizable RPA Rules:

Ensure that the RPA system allows for customizable rules that can be updated as pricing strategies and product offerings evolve.

Integration with Data Sources:

Integrate the RPA system with internal and external data sources to pull in real-time data for more accurate and dynamic pricing.

Monitoring and Analytics:

Integrate monitoring tools to track the performance of RPA bots and use analytics to identify opportunities for further efficiency gains.

Define Boundary for Case Bid Size

Current Proposal:

Set clear thresholds for bid size that determine the level of approval required, minimizing redundant steps in the approval process.

Enhanced Improvements:

Scalable Approval Matrix:

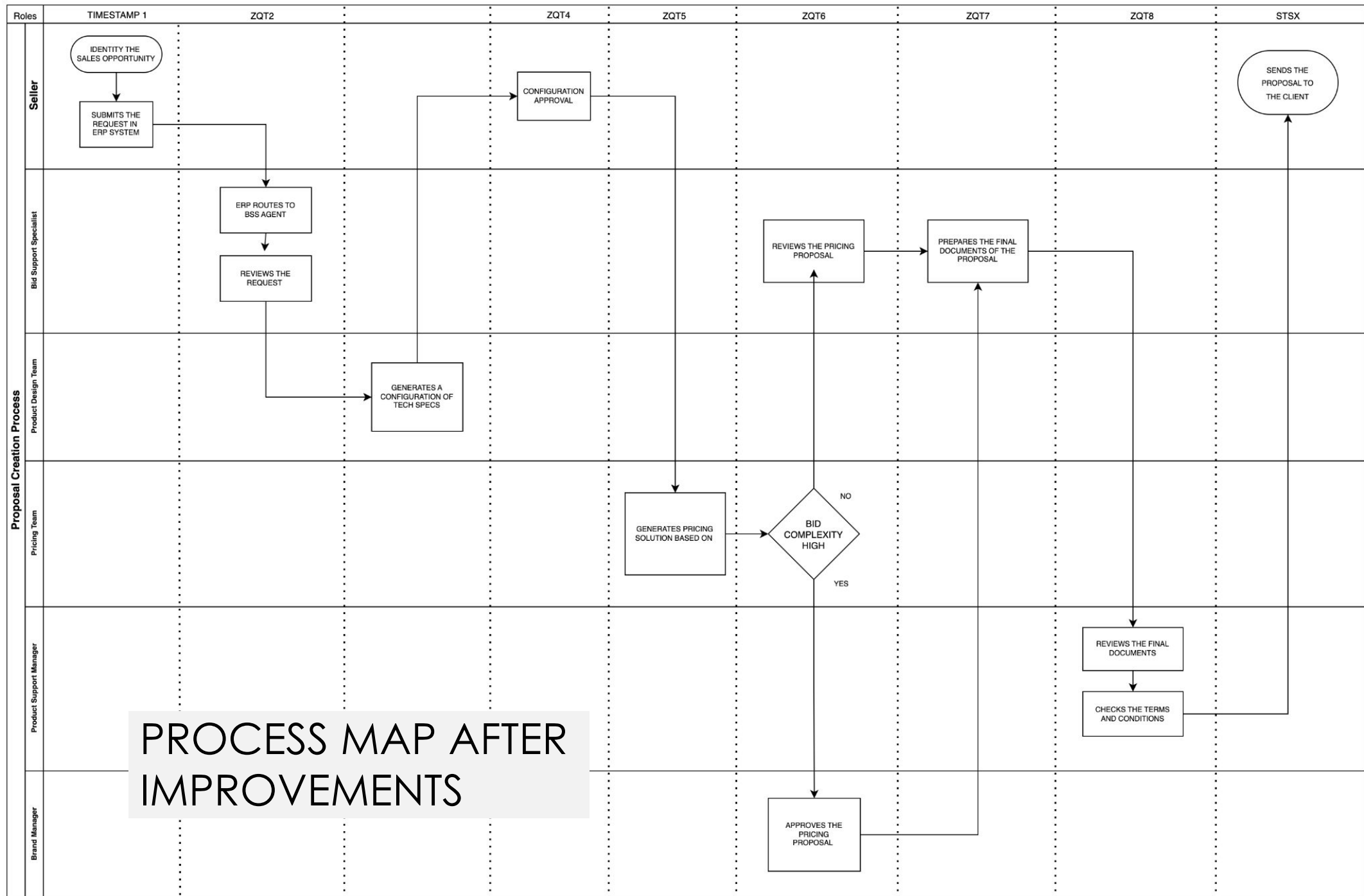
Develop a scalable approval matrix that can adjust the required levels of approval based on various risk factors, not just bid size.

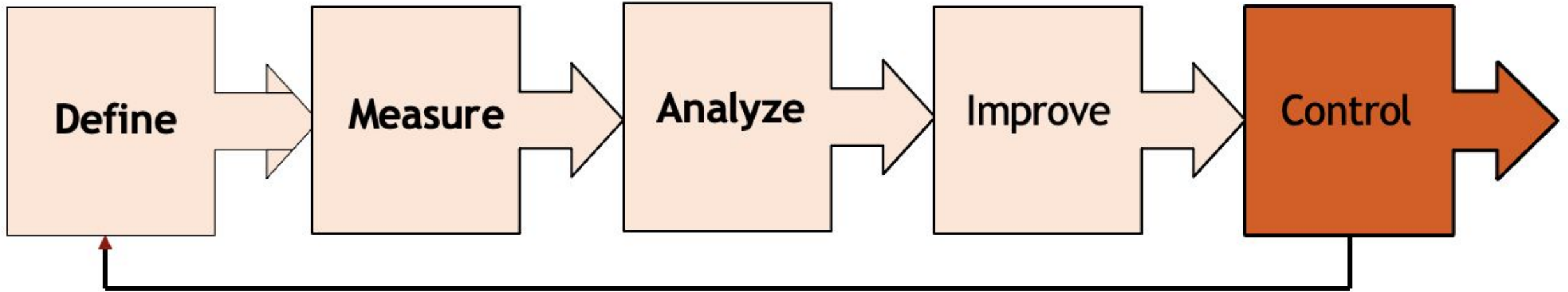
Automated Escalation Paths:

Implement automated escalation paths within the ERP system that trigger additional approvals only when necessary, based on the matrix.

Decision-Making Autonomy:

Empower BSS agents with more decision-making autonomy for lower-risk bids to expedite the process.





CONTROL CHART

Process Component	Control Characteristic	Specification	Measurement Method	Control Method	Reaction Plan
Opportunity Identification & Initial Request	Completeness of information & Timely Submission	All fields completed	ERP timestamp (STS1)	Automated system check	Request sent back to Seller for completion
Request Assignment	Correct BSS assignment	Matched to BSS skillset	ERP timestamp (ZQT2)	Algorithmic matching	Reassignment; refine algorithm
Design & Configuration	Technical specification accuracy	Meets customer requirements	ERP timestamp (ZQT5 & 6)	Design lead review	Rework configuration; log issues
Pricing	Pricing accuracy	Reflects design & within margins	ERP timestamp (ZQT7)	Pricing lead review	Adjust pricing; analyze errors
Approval	Timely approval	Reviewed within 24 hours	ERP approval tracking	Escalation for delays	Notify management for delays
Final Review & Submission	Compliance and accuracy	Adheres to standards and terms	ERP timestamp (ZQT8)	Proposal Support Manager sign-off	Return for correction; address in training
Proposal Delivery to Client	Timely delivery	Delivered within client timeline	ERP timestamp (STSX)	Tracking of delivery times	Analyze cause; address with Seller

THANK YOU

I-MR Control Chart of Average Cycle Time Eliminating Special Cause Variation (Defects):

Assumptions:

- Removing every transaction with average cycle time greater than 35 days from the control chart which is considered as defects.
- The special cause variations are defined as defects in this case.
- Any other RED points seen on the control chart are considered to be a common cause variation since the avg cycle time is less than 35 days.
- These are trial control limits, a new control chart with final limits would be calculated after process improvement recommendations have been implemented.

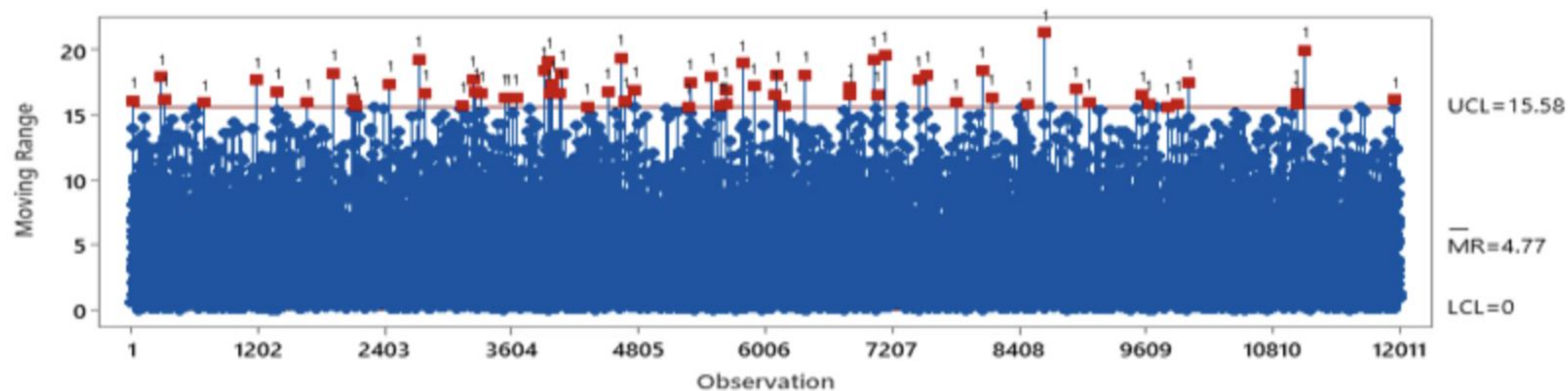
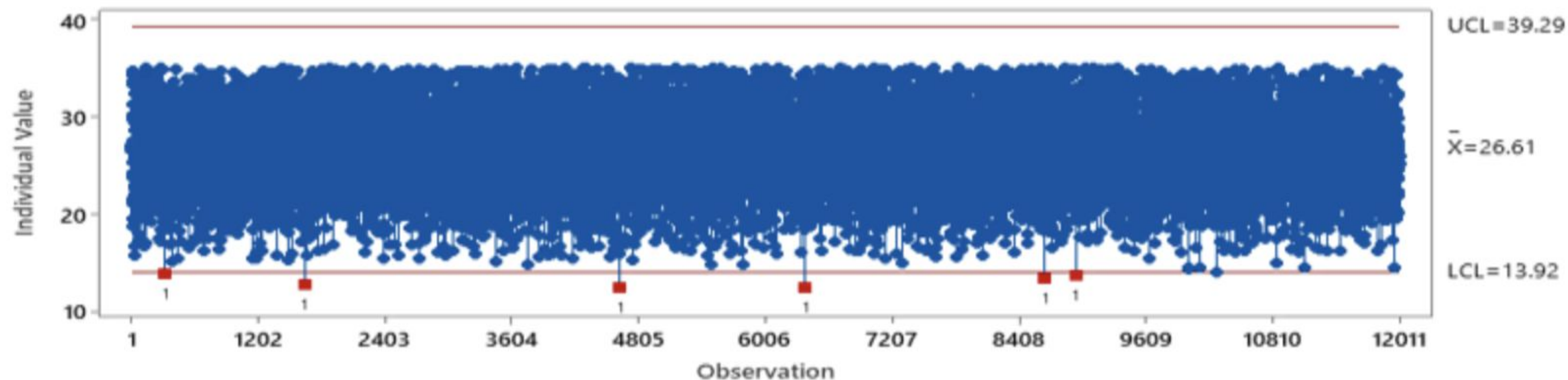
Observations: 53920 (non defective)

Mean, $\bar{x}$ - 26.37 SD, σ = 4.31

UCL = 39.29, LCL = 13.92

mean Moving Range = 4.77

I-MR Chart of CYCLE TIME



Subgroups (observations) to display reduced to the last 500 transactions.

